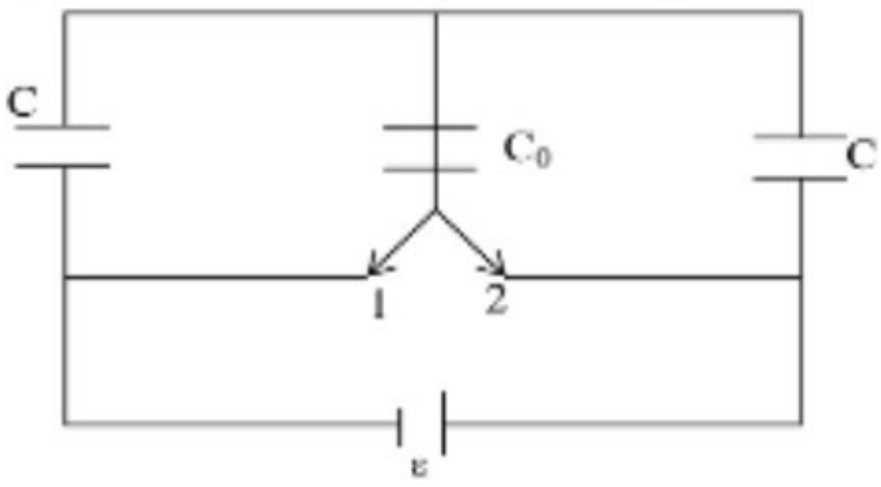
PHYSICS

1) What amount of heat will be generated if switch is shifted from position 1 to position 2



(1) $\frac{CC_0\varepsilon^2}{(C + C_0)}$

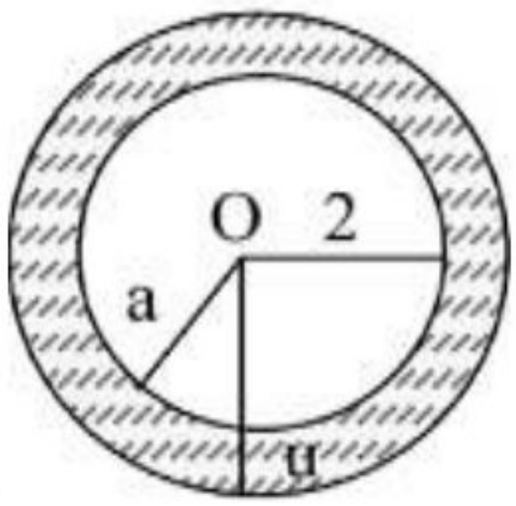
(2) $\frac{CC_0\varepsilon^2}{(2C + C_0)}$

(3) $\frac{2CC_0\varepsilon^2}{(2C + C_0)}$

(4) $\frac{2CC_0\varepsilon^2}{(C + C_0)}$

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2) A point charge q is located at the centre O of a spherical uncharged conducting sphere provided with small orifice. Find the work done to remove the charge to infinity.



(1) $\frac{q^2}{\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$

(2) $\frac{q^2}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$

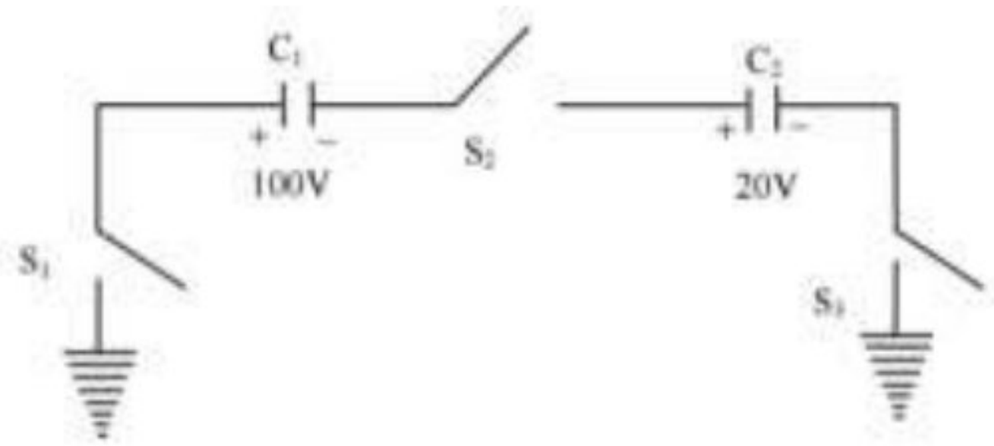
(3) $\frac{q^2}{8\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$

(4) $\frac{4q^2}{\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$

3) In the circuit shown in figure $C_1 = 1\mu F$ and $C_2 = 2\mu F$. Capacitor C_1 is charged to 100 V and C_2 is charged to 20V. After charging they are connected as shown. When the switches S_1, S_2 and S_3 all are

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closed. Find the charge flown through S_1.

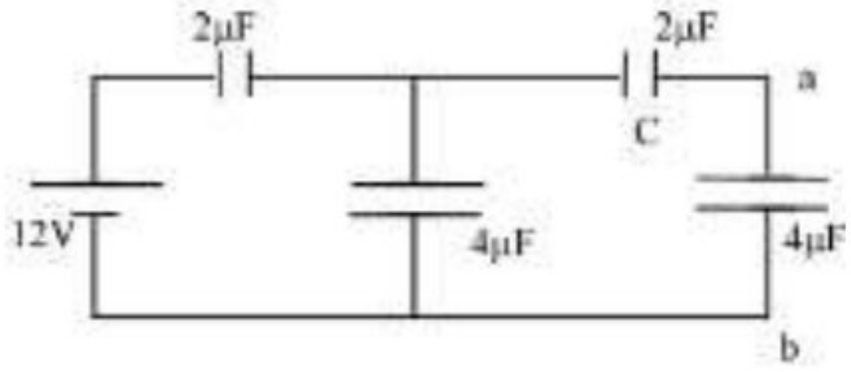
(1) 40μC

(2) 60μC

(3) 80μC

(4) 100μC

4) Find the potential difference $V_a - V_b$ between the points a and b shown in figure



(1) $\frac{11}{13}$V

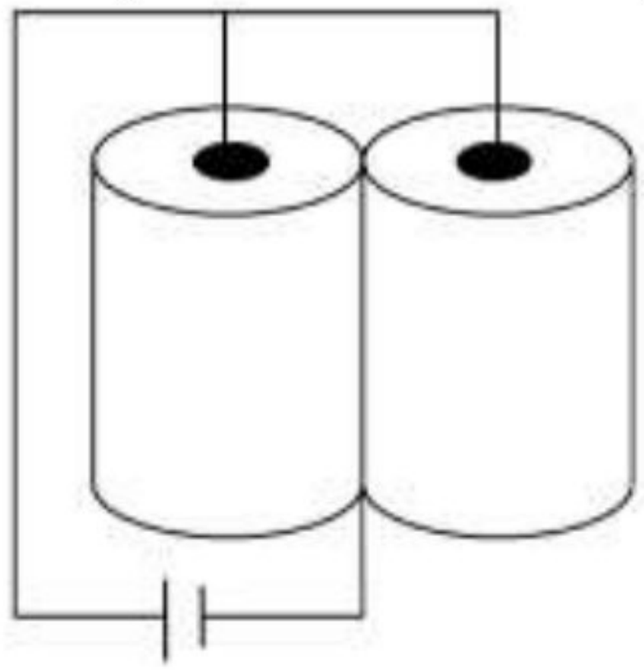
(2) $\frac{12}{13}$V

(3) $\frac{11}{15}$V

(4) $\frac{12}{11}$V

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5) The outer cylinder of two cylindrical capacitors of capacitance 2.2 μF each, are kept in contact and the inner cylinders are connected through a wire. A battery of emf 10V is connected as shown in the figure. Find the total charge supplied by the battery to the inner cylinders.



(1) 44μC

(2) 22μC

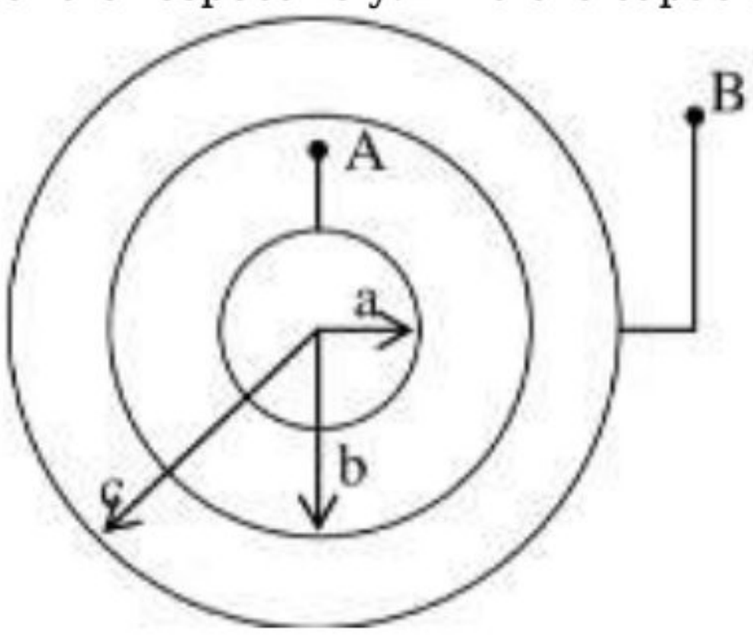
(3) 11μC

(4) 33μC

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6) Consider an assembly of three concentric conducting shells as shown in the figure, of radii a, b and c respectively. Find the capacitance of the assembly between the points A & B.



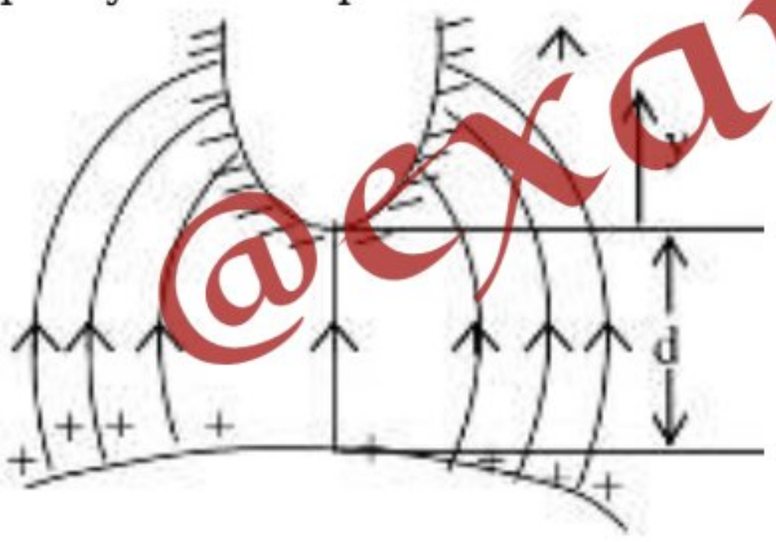
(1) $\frac{3\pi\varepsilon_0 ac}{c-a}$

(2) $\frac{2\pi\varepsilon_0 ac}{c-a}$

(3) $\frac{4\pi\varepsilon_0 ac}{c-a}$

(4) $\frac{8\pi\varepsilon_0 ac}{c-a}$

7) Two conductor carrying equal and opposite charge create non-uniform field as shown in the figure what is the capacity of this capacitor is the field varies along the y axis as $E = \frac{Q}{\varepsilon_0 A}\left[1 + by^2\right]$, where



is a constant.

(1) $C = \frac{A\varepsilon_0}{d\left[1 + \frac{1}{7}Bd^2\right]}$

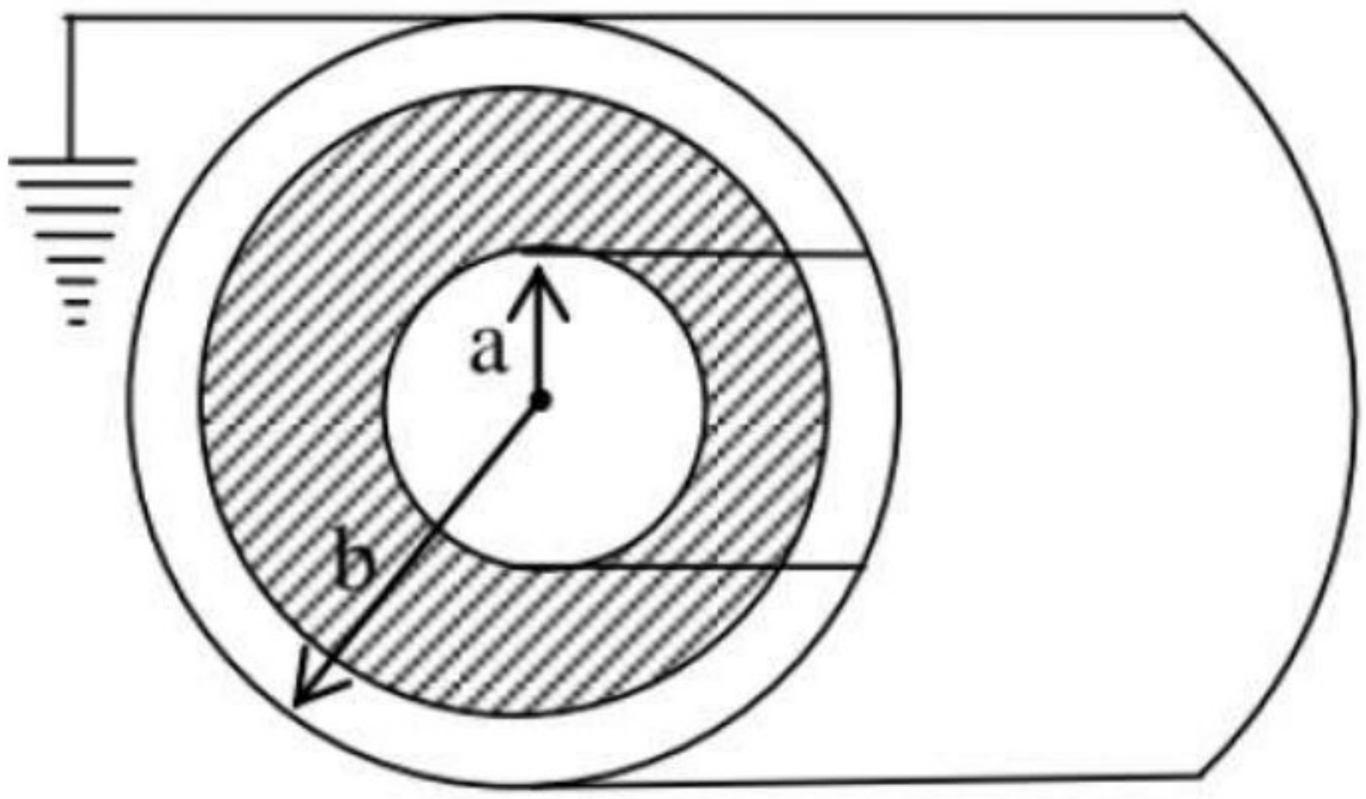
(2) $C = \frac{A\varepsilon}{d\left[1 + \frac{1}{2}Bd^2\right]}$

(3) $C = \frac{A\varepsilon_0}{d\left[1 + \frac{1}{3}Bd^2\right]}$

(4) $C = \frac{A\varepsilon_0}{d\left[1 + \frac{1}{5}Bd^2\right]}$

8) A co-axial cable consists of inner solid cylinder of radius a and outer coaxial pipe of inner radius b. the outer conducting pipe is earthed. Calculate the capacitance between the conductors, if the inner cylinder carries charged λ per unit length and the length of the cable is L.

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(1) $C = \frac{2\pi\varepsilon_0 L}{\ln\left(\frac{b}{a}\right)}$

(2) $C = \frac{\pi\varepsilon_0 L}{\ln\left(\frac{b}{a}\right)}$

(3) $C = \frac{4\pi\varepsilon_0 L}{\ln\left(\frac{b}{a}\right)}$

(4) $C = \frac{\pi\varepsilon_0 L}{2\ln\left(\frac{b}{a}\right)}$

9) A total charge Q flows across a resistor R during a time interval T in such a way that the current vs time graph for $0 \rightarrow T$ is like the loop of a sin curve in the range $0 \rightarrow \pi$. What will be the total heat generated in the resistor.

(1) $\frac{Q^2\pi^2 R}{4T}$

(2) $\frac{Q^2\pi^2 R}{8T}$

(3) $\frac{Q^2\pi^2 R}{2T}$

(4) $\frac{Q^2\pi^2 R}{6T}$

10) One end of a Nichrome wire of length 2 L and cross-sectional area A is attatched to an end of another Nichrome wire of length L and cross-sectional area 2 A. If the free end of the longer wire is at an electric potential of 8.0 volts, and the free end of the shorter wire is at an electric potential of 1.0 volt, the potential at the junction of the two wires is equal to

(1) 2.4 V

(2) 3.2 V

(3) 4.5 V

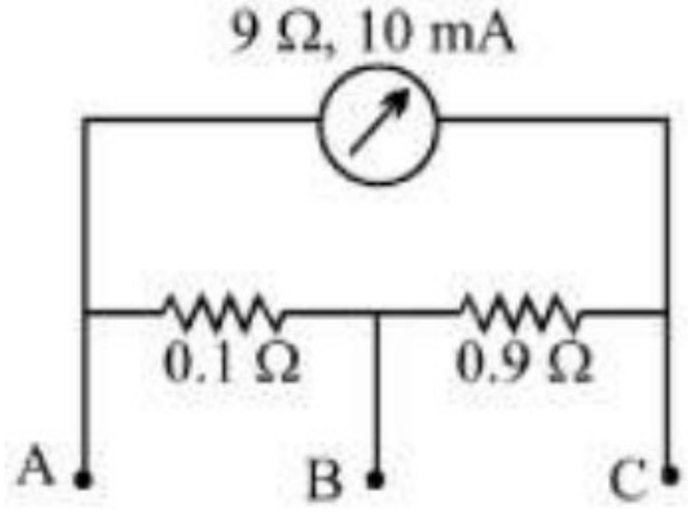
(4) 5.6 V

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11) A milliammeter of range 10 mA and resistance 9Ω is joined in a circuit as shown. The metre gives full-scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B(C is left isolated). The value of I is



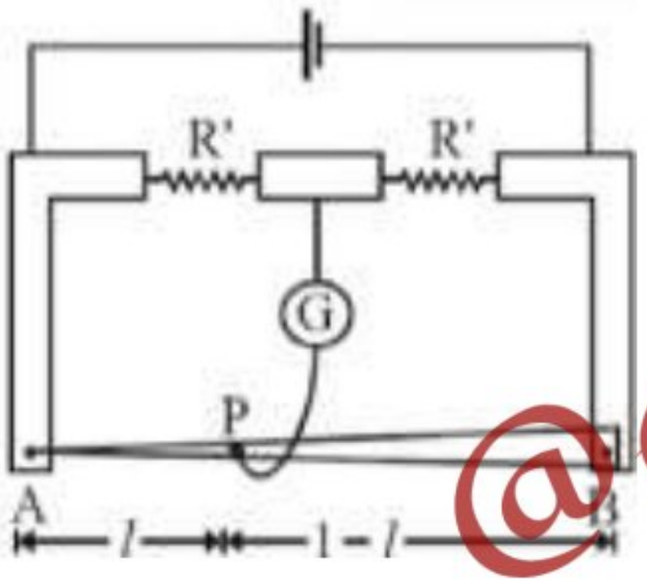
(1) 100 mA

(2) 900 mA

(3) 1 A

(4) 1.1 A

12) A meter bridge, the wire of length 1 m has a non-uniform cross-section such that, the variation $\frac{dR}{d\ell}$ of its resistance R with length ℓ is $\frac{dR}{d\ell} \propto \frac{1}{\sqrt{\ell}}$ two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P. what is the length AP?



(1) 0.2 m

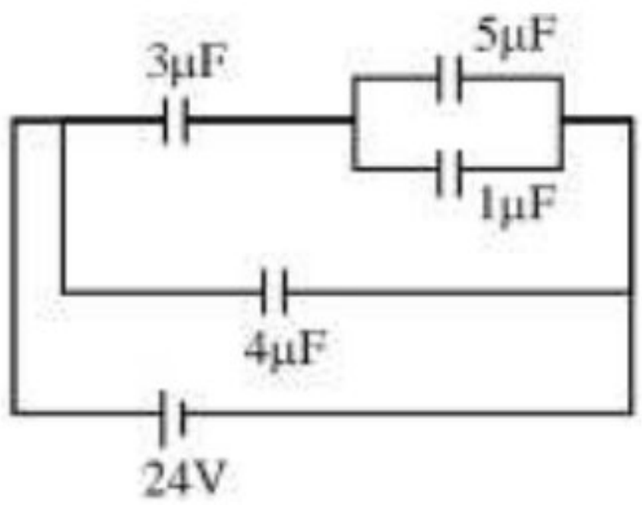
(2) 0.35 m

(3) 0.25 m

(4) 0.3 m

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13) In the circuit shown, the energy stored in 1μF capacitor is



(1) 40 μJ

(2) 64 μJ

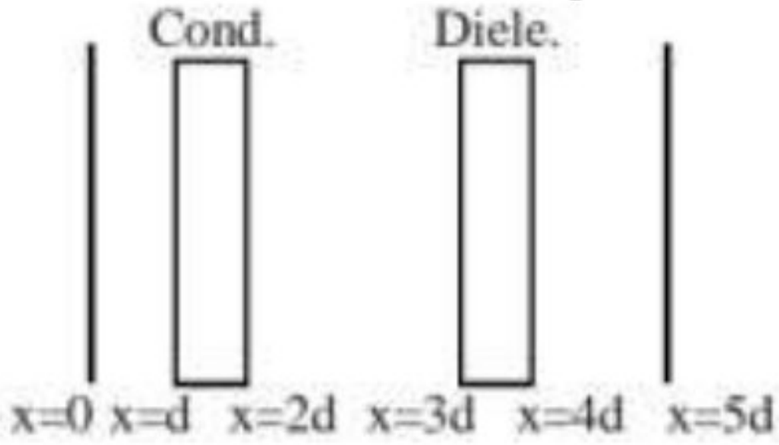
(3) 32 μJ

(4) None

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14) The distance between plates of a parallel plate capacitor is 5d. Let the positively charged plate is at x = 0 and negatively charged plate is at x = 5d. Two slabs one of conductor and other of a dielectric of equal thickness d are inserted between the plates as shown in figure. Potential versus distance graph will look like:



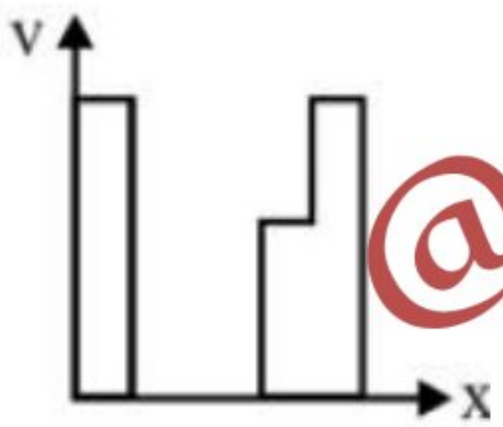
(1)



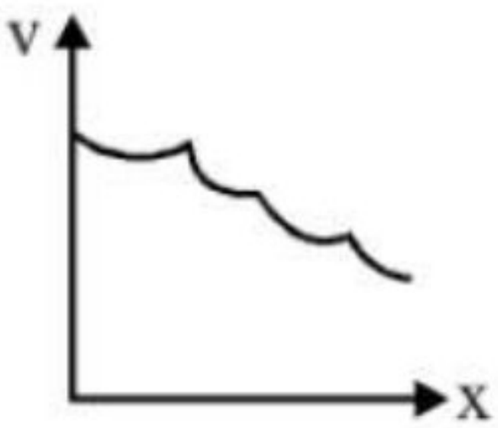
(2)



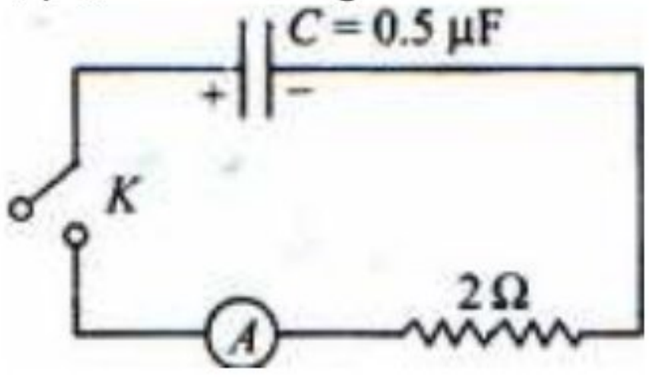
(3)



(4)



15) A charged capacitor is allowed to discharge through a resistor by closing the key at the instant t = 0, At the instant t = (ln 4) μs, the reading of the ammeter falls half the initial value. The resistance of the ammeter is equal to



(1) 1 MΩ

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(2) 1Ω

(3) 2 Ω

(4) 2MΩ

16) A capacitor of capacitance C is charged by a battery whose internal resistance is R. The time after which potential difference across resistor becomes n times to that across capacitor is

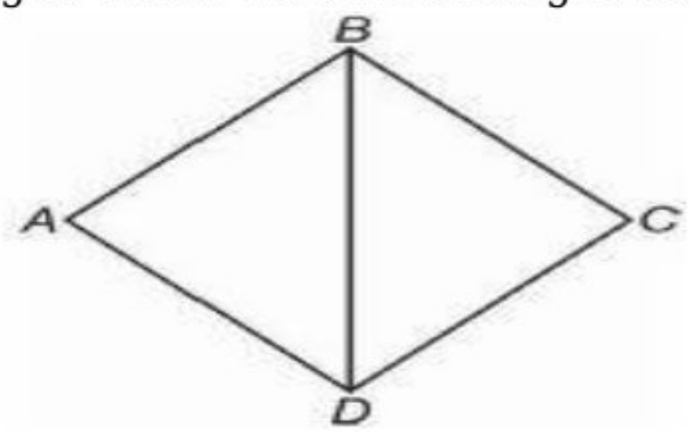
(1) $RC\ell n\left(\frac{1+n}{n}\right)$

(2) $RC\ell n\left(\frac{n}{1+n}\right)$

(3) $RC\ell n\left(\frac{n}{n-1}\right)$

(4) $RC\ell n\,(1+n)$

17) A uniform wire of resistance R stretched uniformly to n times and then cut to form five wires of equal length. These wire are arranged as shown in the fig. The effective resistance between point A and B is:



(1) $\frac{nR}{5}$

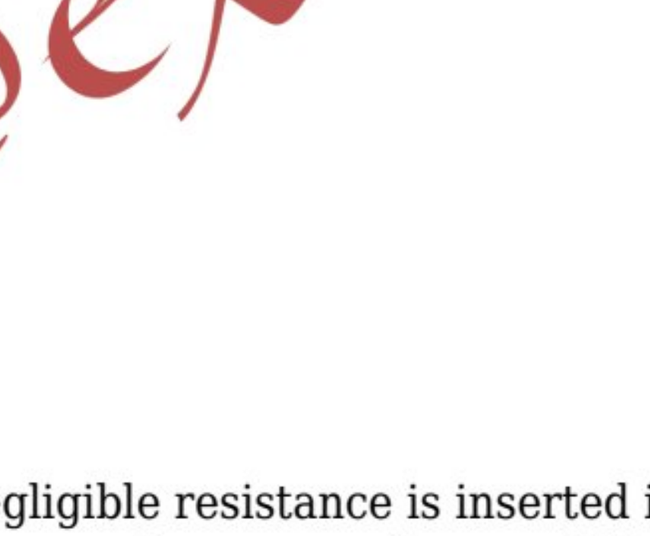
(2) $\frac{R}{(5n)}$

(3) $\frac{n^2R}{5}$

(4) $\frac{R}{(5n^2)}$

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18) When an ammeter of negligible resistance is inserted in series with circuit it reads 1 A. When the voltmeter of very large resistance is connected across X it reads 1V. When the points A and B are shorted by a conducting wire, the voltmeter measures 10 V across the battery. The internal resistance of the battery is:



(1) zero

(2) 0.2 Ω

(3) 0.5 Ω

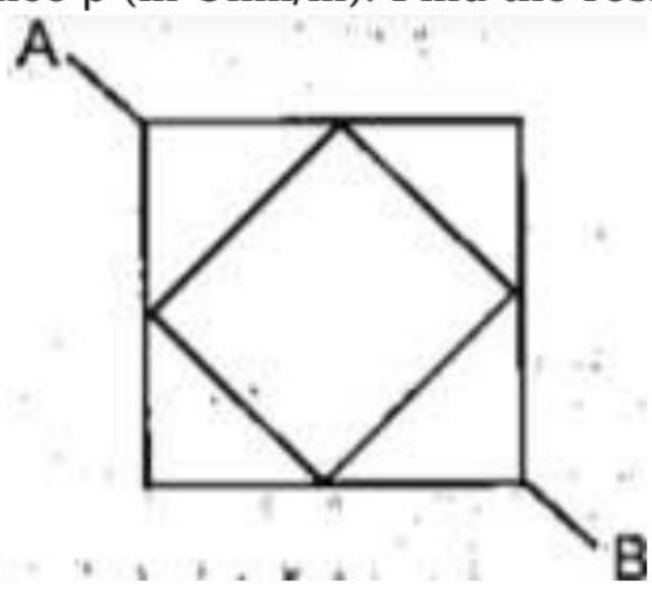
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(4) 0.1 Ω

19) A wire has linear resistance ρ (in Ohm/m). Find the resistance R between points A and B if the side of the "big" square is d:



(1) $\frac{\rho d}{\sqrt{2}}$

(2) $\sqrt{2}\rho d$

(3) 2ρd

(4) None of these

20) The wattage rating of a light bulb indicates the power dissipated by the bulb if it is connected across 110 V DC potential difference . If a 50 W and 100 W bulb are connected in series to a 110 V DC source, how much power will be dissipated in the 50 W bulb:

(1) 50 W

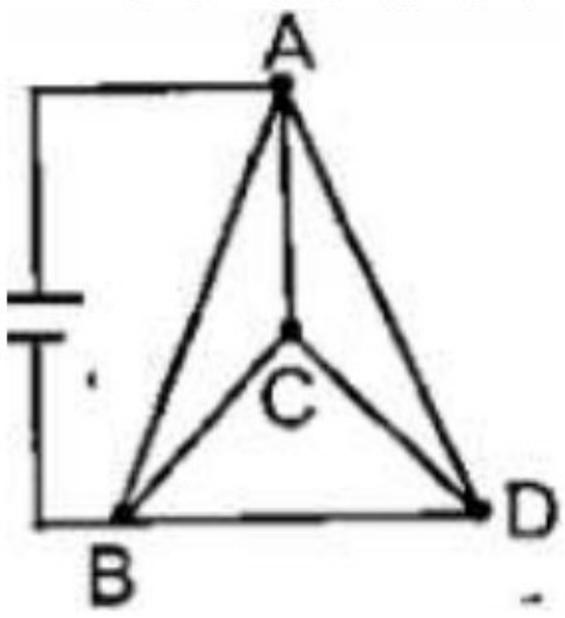
(2) 100 W

(3) 22 W

(4) 11 W

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21) A tetrahedral is consisting of 6 identical wires as shown in figure. Each wire is having a resistance of 2Ω. When an ideal cell of emf 2 V is connected across AB, as shown then the current through CD is:



(1) 1 A

(2) $\frac{6}{19}$A

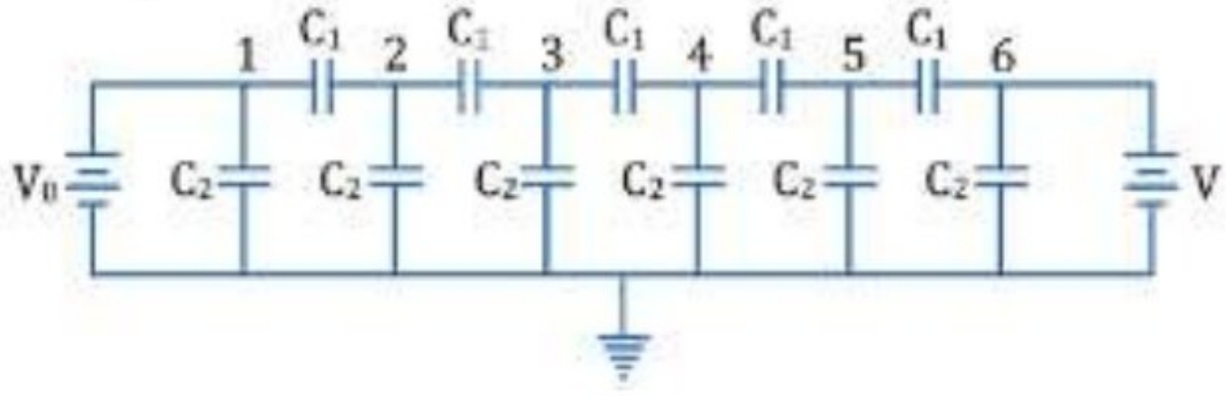
(3) 4 A

(4) zero

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22) In the circuit shown, ratio of capacitances is $\frac{C_2}{C_1} = \frac{4}{3}$ and terminal voltage of the battery connected at the left end is V_o . Given that the potentials of nodes 1, 2, 3, 4, 5 and 6 are in geometric progression, when the terminal voltage of the battery connected at the right most end is V. (In



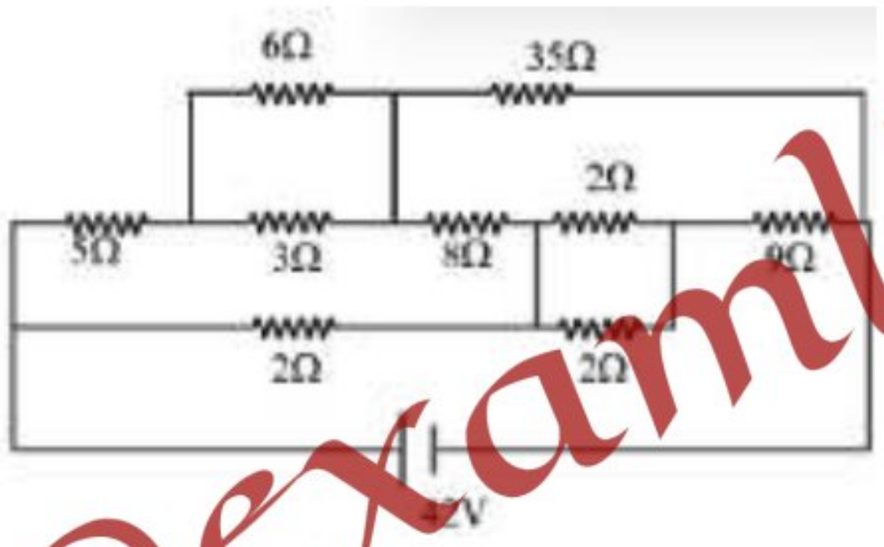
steady state condition) Then :-

(1) V may be $\frac{V_o}{3^5}$

(2) V may be V

(3) V may be $\frac{V_o}{4}$

(4) V cannot be found with this data



23) In given circuit:

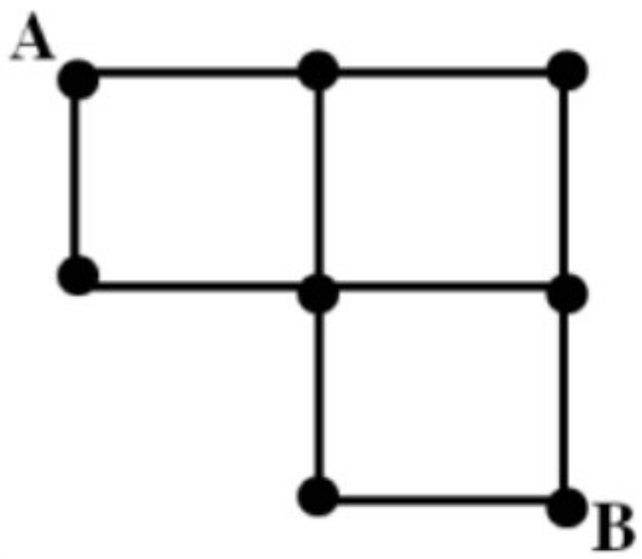
(1) Power dissipated in 5Ω resistor is 10 W.

(2) Power dissipated in 5Ω resistor is 4 W.

(3) Power dissipated in 8Ω resistor is 0 W

(4) Power dissipated in 8Ω resistor is 2 W

24) Find equivalent resistance in ohm between points A and B for the circuit shown in drawing if



resistance of each link is equal to 1 Ω.

(1) 1.71

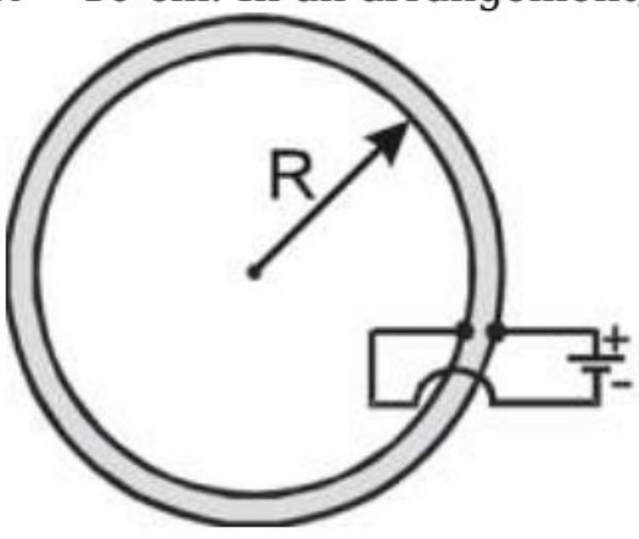
(2) 2.51

(3) 4

(4) 1.25

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25) A spherical shell, made of material of electrical conductivity $\frac{10^9}{\pi}(\Omega - m)^{-1}$, has thickness t = 2 mm and radius R = 10 cm. In an arrangement, its inside surface is kept at a lower potential than its outside surface.



The resistance offered by the shell is equal to n × 10^{-11} Ω. Find n

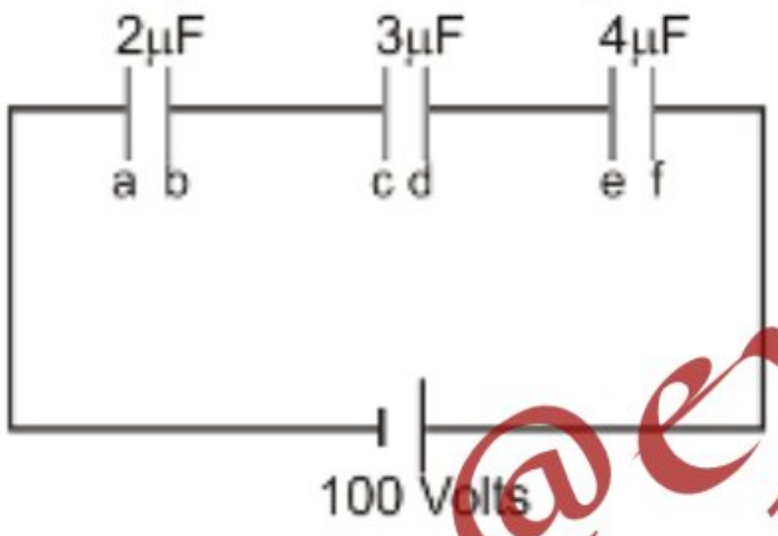
(1) 3

(2) 5

(3) 7

(4) 9

26) Three capacitors are connected to D.C. source of 100 volts shown in the adjoining figure. If the charge accumulated on plates of C_1, C_2 and C_3 are q_a, q_b, q_c, q_d, q_e and q_f respectively, then



(1) $q_b + q_d + q_f = \frac{100}{9}C$

(2) $q_b + q_d + q_f = 0$

(3) $q_a + q_c + q_e = 50C$

(4) $q_b = q_d = q_f$

27) Six capacitors each of capacitance ' C ' are connected as shown in the figure and initially all the capacitors are uncharged. Now a battery of emf = ε is connected. How much charge will flow through the battery if the switch is on:

(1) $\frac{9C_\varepsilon}{5}$

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(2) $\frac{11C_E}{5}$

(3) $\frac{13C_E}{5}$

(4) $\frac{7C_E}{5}$

28) A capacitor of capacitance C_0 is charged to a potential V_0 and then isolated. A capacitor C is then charged from C_0, discharged and charged again; the process being repeated n times. Due to this, potential of the larger capacitor is decreased to V value of C is:

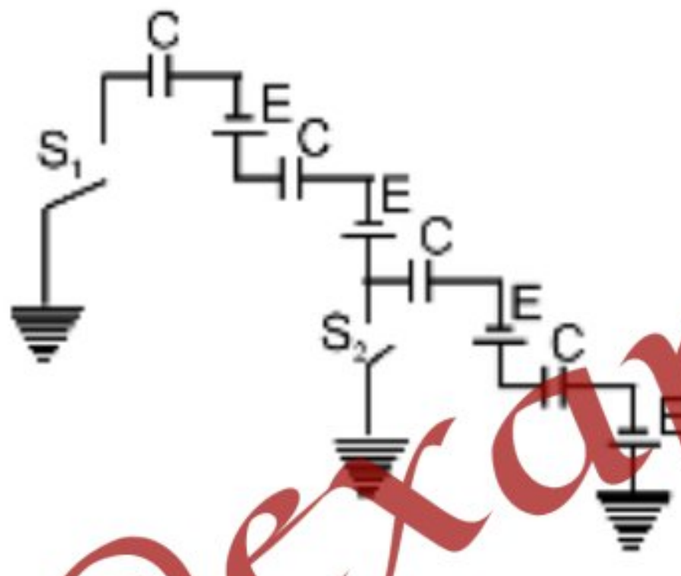
(1) $C_0\ [V_0/V]1/n$

(2) $C_0[(V_0/V)^{1/n} - 1]$

(3) $C_0[(V_0/V) - 1]$

(4) $C_0\ [(V/V_0)^n + 1]$

29) In the given circuit, all the capacitors are initially uncharged. After closing the switch S_1 for a long time suddenly S_2 is also closed and kept closed for a long time. Total heat produced after closing S_2 will be :



(1) $4\ C\varepsilon^2$

(2) $1/2\ C\varepsilon^2$

(3) $2\ C\varepsilon^2$

(4) 0

30) Three long concentric conducting cylindrical shells have radii R, 2R and $2\sqrt{2}R$. Inner and outer shells are connected to each other. The capacitance across middle and inner shells per unit length is:

(1) $\frac{\frac{1}{3}\varepsilon_0}{\ln 2}$

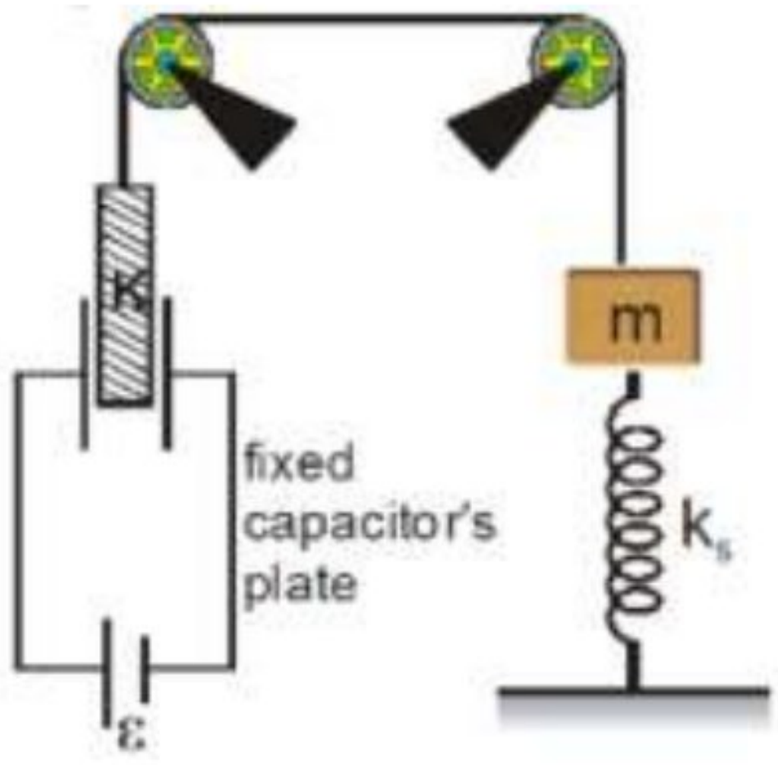
(2) $\frac{6\pi\varepsilon_0}{\ln 2}$

(3) $\frac{\pi\varepsilon_0}{2\ln 2}$

(4) None

31) Initially the block of mass m is in equilibrium in vertical plane as shown and dielectric slab of some mass. When it is slightly displaced from mean position (pulleys are ideal and spring is

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massless):

Dielectric will perform SHM having time period

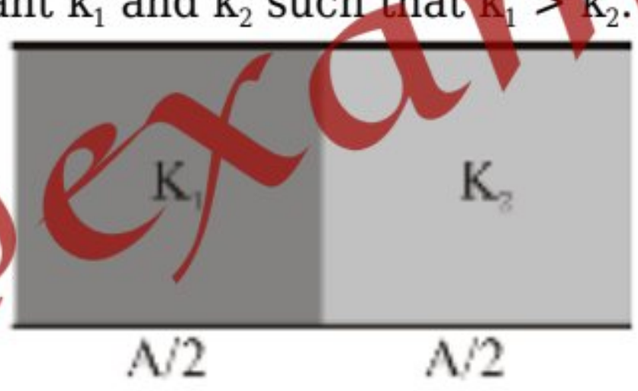
(1) $2\pi\sqrt{\frac{m}{k_s}}$

(2) Dielectric will perform SHM having time period more than $2\pi\sqrt{\frac{m}{k_s}}$

(3) Dielectric will perform SHM having time period less than $2\pi\sqrt{\frac{m}{k_s}}$

(4) Dielectric will not perform periodic motion.

32) Inside a parallel plate capacitor of capacitance C (without dielectrics) dielectrics are filled which have dielectric constant k_1 and k_2 such that $k_1 > k_2$. The capacitor is now connected to a battery of voltage V then :-



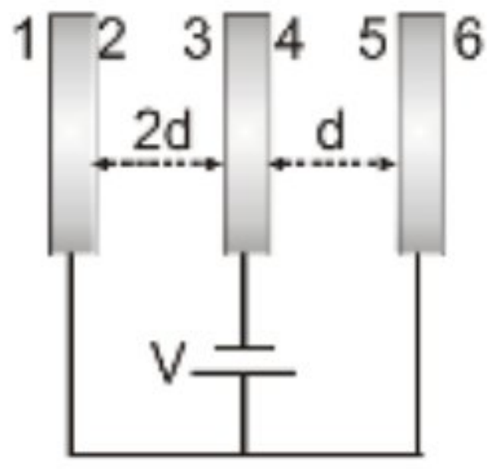
(1) New capacitance of the capacitor will be $\left(\frac{k_1 k_2}{k_1 + k_2}\right) C$

(2) Field in dielectric 1 will be more than the field in dielectric 2

(3) Surface charge density on the capacitor plates will be uniform

(4) Field in both the dielectric regions will be the same.

33) In the figure shown, area of each plate is A. Match the following:



Column - I	Column -II

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(A)	Charge on plate 3	**(P)**	Zero
(B)	Charge on plate 5	**(Q)**	V
(C)	Potential difference between plates 2 and 3	**(R)**	$\frac{\varepsilon_0 A}{2d}V$
(D)	Potential difference between plates 2 and 5	**(S)**	$\frac{\varepsilon_0 A}{d}V$
		(T)	None of these

(1) (A) T, (B) S, (C) Q, (D) P

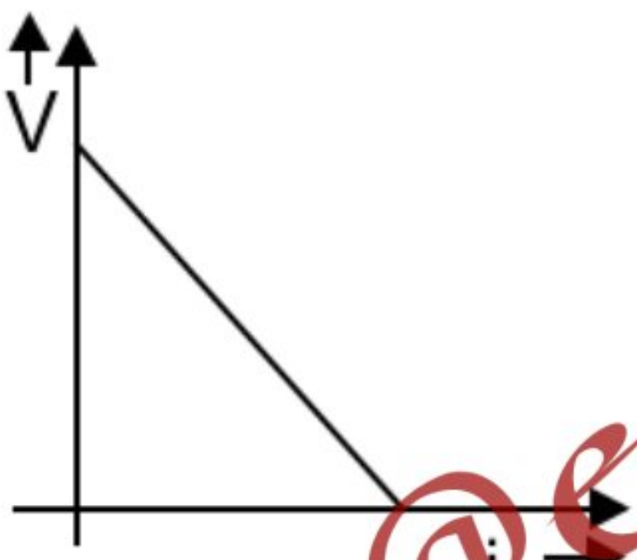
(2) (A) P, (B) Q, (C) T, (D) S

(3) (A) T, (B) Q, (C) S, (D) P

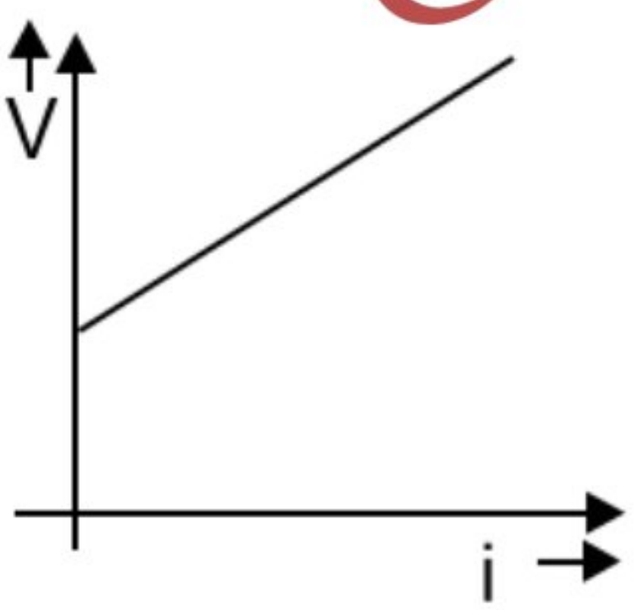
(4) (A) P, (B) S, (C) Q, (D) T

34) If internal resistance of a cell is proportional to current drawn from the cell. Then the best representation of terminal potential difference of a cell with current drawn from cell will be:

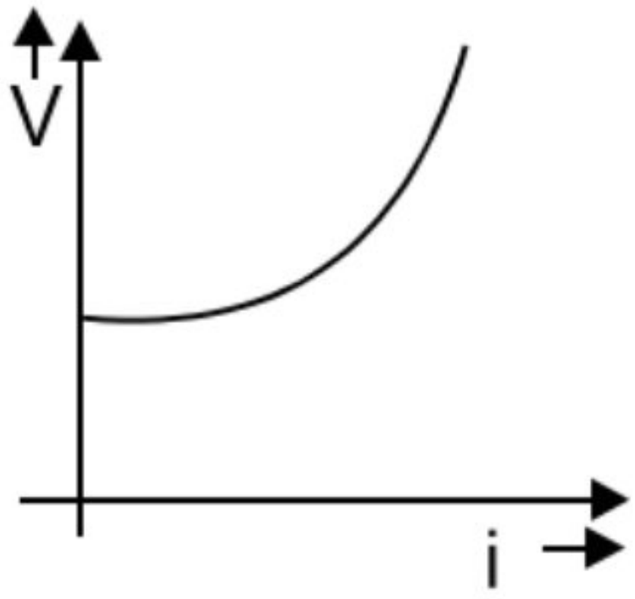
(1)



(2)



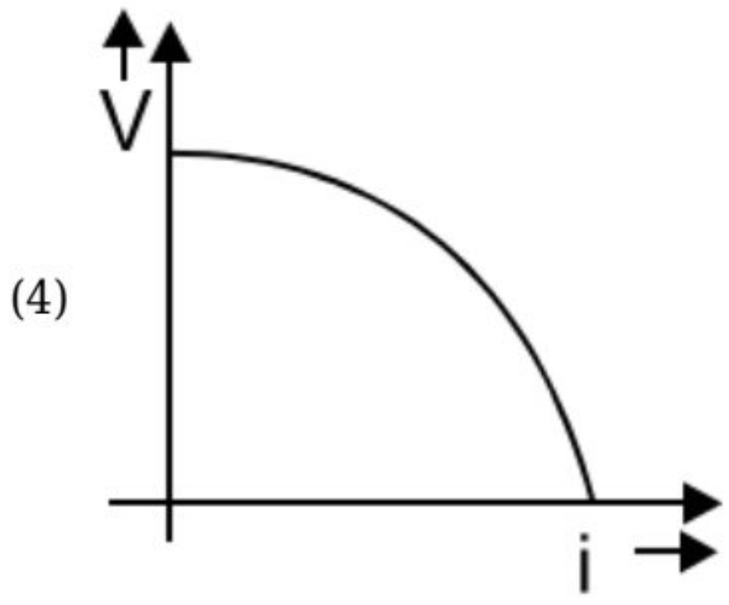
(3)



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(4)

35) The space between two concentric metallic spheres of radii R_1 and R_2 is filled with a material of conductivity σ. The total resistance of the system is :-

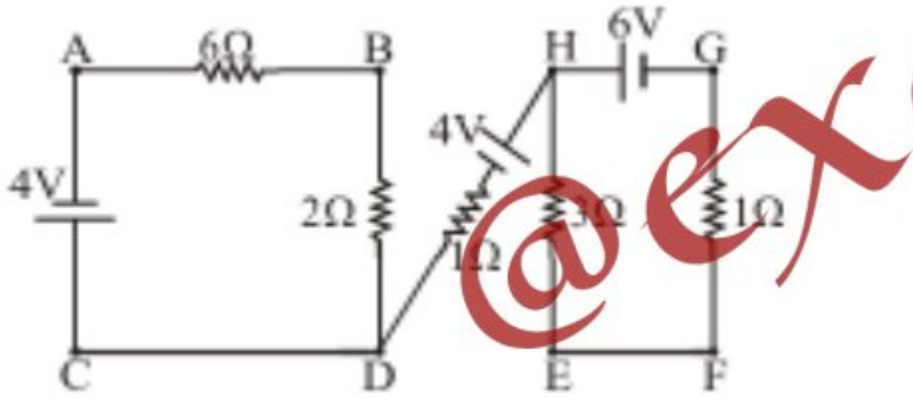
(1) $\frac{R_2 - R_1}{4\pi R_1 R_2 \sigma}$

(2) $\frac{R_2 + R_1}{4\pi R_1 R_2 \sigma}$

(3) $\frac{R_2 + R_1}{4\pi R_1 R_2 \sigma^2}$

(4) $\frac{R_2 - R_1}{R_1 R_2}$

36) In the network shown in the figure, choose the incorrect statement:



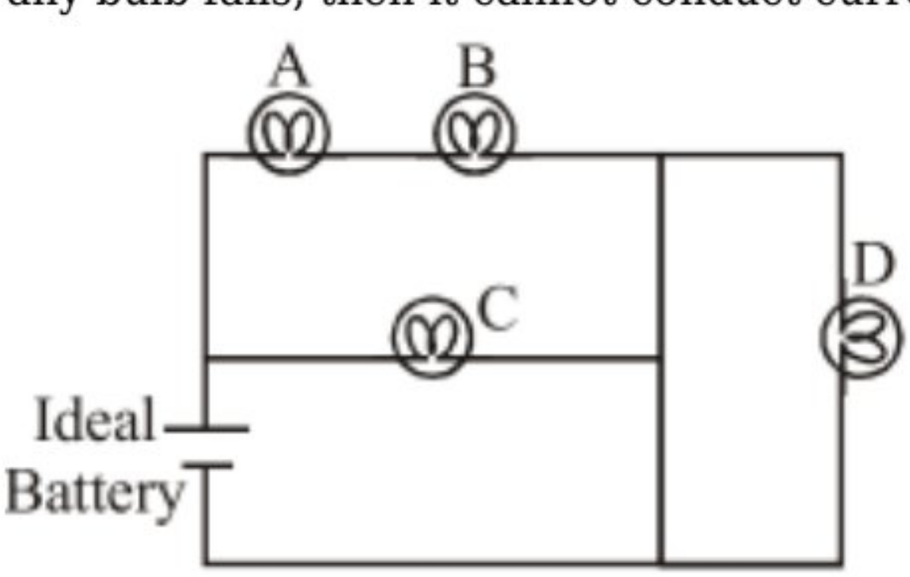
(1) Potential difference between points H and B is 5 V

(2) Potential difference between points D and B is 1 V

(3) Potential difference between points D and E is 1 V

(4) Potential difference between points D and G is 2 V

37) Four identical bulbs A, B, C, D are connected in a circuit as shown in figure. Now whenever any bulb fails, then it cannot conduct current through it. Then choose incorrect statement:



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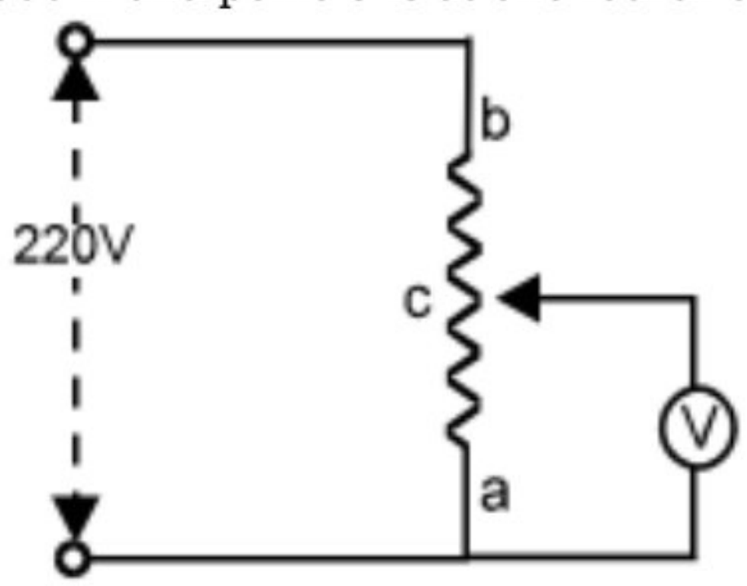
(1) Brightness of bulb C is highest

(2) If C fails, brightness of bulb D increases

(3) If C fails, brightness of all bulbs remains same

(4) If A fails, B will not glow

38) A potential difference of 220 volts is maintained across a 12000 Ω rheostat as shown in fig. The voltmeter V has a resistance of 6000 Ω and point C is at one fourth of the distance from a to



b. What is the reading of voltmeter?

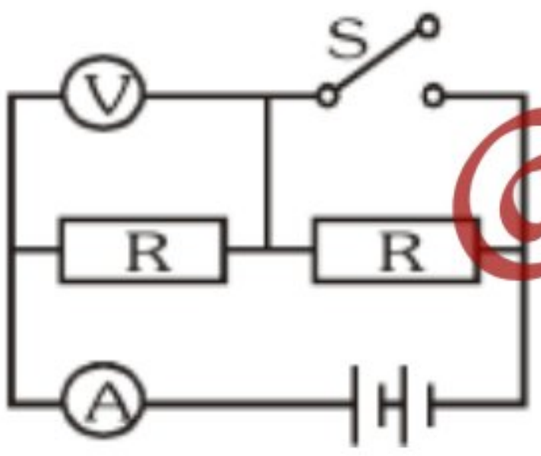
(1) 50 V

(2) 60 V

(3) 40 V

(4) None

39) The diagram shows a circuit with two identical resistors. The battery has a negligible internal resistance. What will the effect on the ammeter and voltmeter be if the switch S is closed?



	Column I		**Column II**
(A)	Ammeter reading	**(P)**	Increases
(B)	Voltmeter reading	**(Q)**	Decreases
(C)	Equivalent resistance of circuit	**(R)**	Does not change
(D)	Power dissipated across R in right branch	**(S)**	Becomes zero

(1) (A) → (P) ; (B) → (P) ; (C) →(Q) ; (D) →(Q,S)

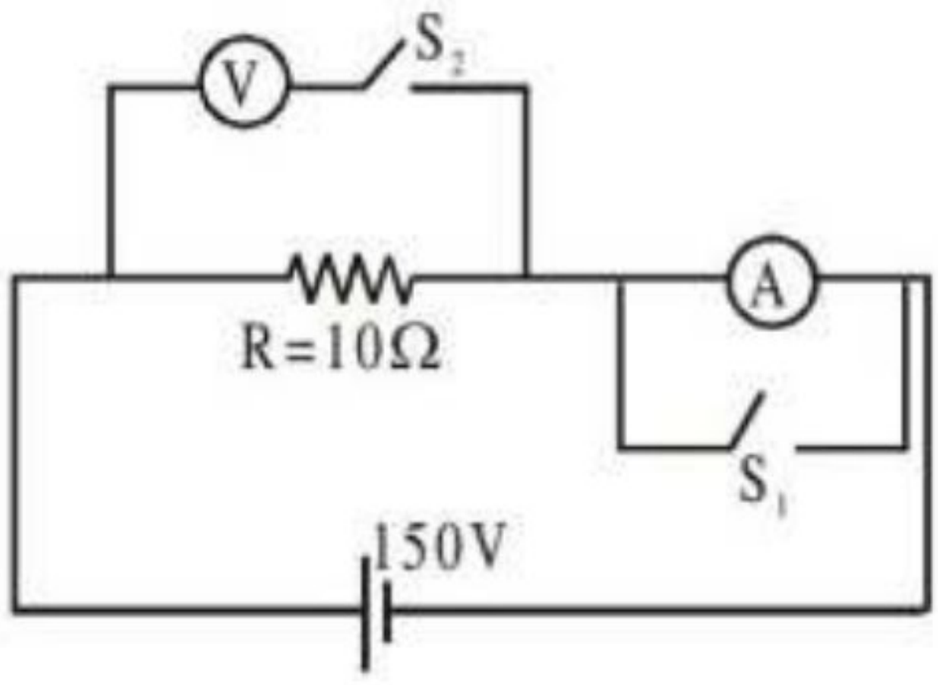
(2) (A) → (Q) ; (B) → (P) ; (C) →(R) ; (D) →(P,S)

(3) (A) → (S) ; (B) → (R) ; (C) →(P) ; (D) →(R,Q)

(4) (A) → (R) ; (B) → (R) ; (C) →(S) ; (D) →(R,S)

40) Voltmeter of resistance 10 Ω and ammeter of resistance 10 Ω are attached in the following

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circuit with switches S_1 and S_2

	Column I (switch position)		**Column II (reading of voltmeter & ammeter)**
(P)	S_1 open, S_2 closed	**(1)**	0, 7.5 A
(Q)	S_1closed, S_2 open	**(2)**	50 V, 10 A
(R)	S_1 open, S_2 open	**(3)**	0, 0
(S)	S_1 closed, S_2 closed	**(4)**	150 V, 0

(1) (A) → (P); (B) → (Q); (C) → (R); (D) → (S)

(2) (A) → (R); (B) → (P); (C) → (S); (D) → (Q)

(3) (A) → (S); (B) → (Q); (C) → (P); (D) → (R)

(4) (A) → (Q); (B) → (R); (C) → (P); (D) → (S)

41) In an RC circuit, the time required for the charge on a capacitor to build up to a given fraction of its equilibrium value, is independent of:

(1) the value of the applied emf to the circuit

(2) the value of C

(3) the value of R

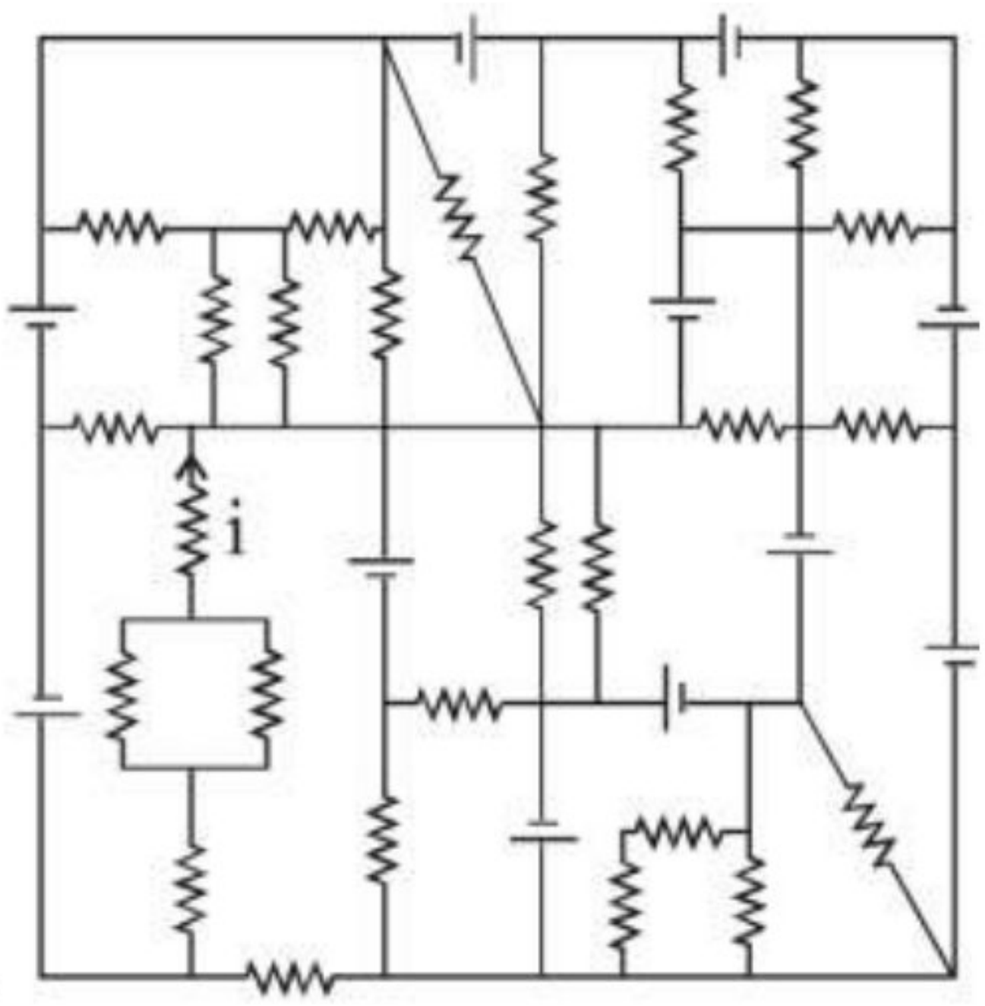
(4) none of the above

42) What is the current i (in A) in figure, where all resistances are 4 Ω and all batteries have an emf

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of 10 V?

(1) 6

(2) 2

(3) 4

(4) 8

43) A' parallel plate capacitor is charged up to voltage V_0 and then detached from the charger. Now, the distance between the plates is varied periodically according to the law $d = d_0 + d_1 \sin \omega t$. The value of d_1 if the potential difference across the capacitor changes according to the law

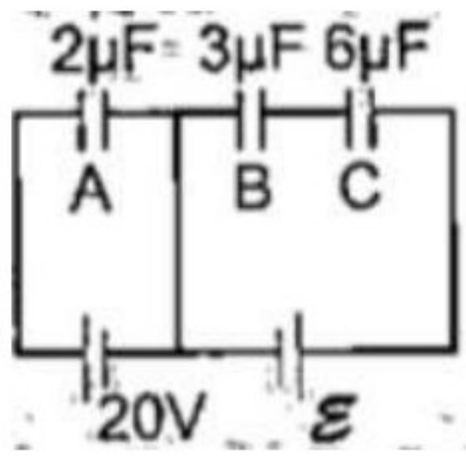
$$V = V_0\left(1 + \frac{1}{2}\sin\omega t\right)$$ is:

(1) d_0

(2) $\frac{d_0}{2}$

(3) $\frac{3d_0}{2}$

(4) $2d_0$



44) In the given circuit if emf ε is increased the steady state charge:

(1) on each capacitor will decrease

(2) will increase for capacitors B and C and decrease for capacitor A

(3) on each capacitor will increase

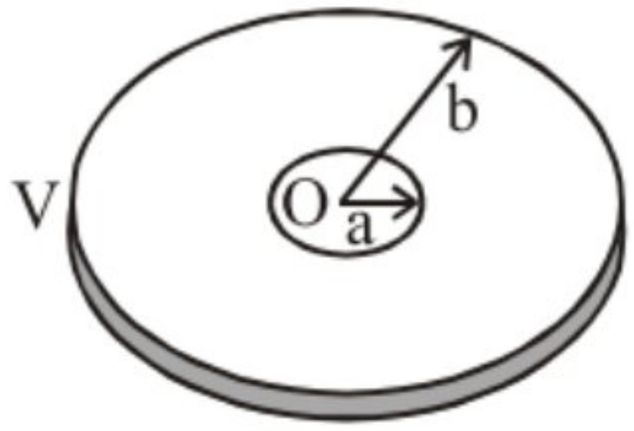
(4) remains constant for A and increases for capacitors B and C

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45) A circular portion is cut out of a disc of thickness t, its resistivity is P and radii of disc are a and b (b > a). A potential difference is maintained between outer and inner cylindrical surfaces of



the disc. What is resistance of the disc?

(1) $\frac{P}{2\pi t}\ln\left(\frac{b}{a}\right)$

(2) $p\left(\frac{1}{a}-\frac{1}{b}\right)$

(3) $2\pi Pt\left(\frac{1}{a^2}-\frac{1}{b^2}\right)$

(4) $\frac{P}{2\pi t}\left(\frac{b^2-a^2}{ab}\right)$

CHEMISTRY

1) When a certain conductivity cell was filled with 0.1 M KCl, it had a resistance of 85 Ω at 25ºC. When the same cell was filled with an aqueous solution of 0.052 M unknown electrolyte, the resistance was 96Ω. Calculate the molar conductivity of the unknown electrolyte at this concentration.
Given :- (Conductivity of 0.1 M KCl = $1.29 \times 10^{-2}\ \Omega^{-1}\ cm^{-1}$)

(1) $120\ \Omega^{-1}\ cm^2\ mol^{-1}$

(2) $219.65\ \Omega^{-1}\ cm^2\ mol^{-1}$

(3) $200.35\ \Omega^{-1}\ cm^2\ mol^{-1}$

(4) $285.2\ \Omega^{-1}\ cm^2\ mol^{-1}$

2) The ionisation constant of a weak acid is 1.6×10^{-5} and the molar conductivity at infinite dilution is $380 \times 10^{-4}\ Sm^2/mol$. If cell constant is $0.01\ m^{-1}$ then conductance of 0.01 M acid solution is :-

(1) 1.52×10^{-5} S

(2) 1.52 S

(3) 1.52×10^{-3} S

(4) 1.52×10^{-4} S

3) **Statement-I :** The limiting molar conductivity of KCl (strong electrolyte) is higher compared to that of CH_3COOH (weak electrolyte).
Statement-II : Molar conductivity increases with decrease in concentration of electrolyte.

(1) Statement-I is true but Statement-II is false.

(2) Statement-I is false but Statement-II is true.

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(3) Both Statement-I and Statement-II are true.

(4) Both Statement-I and Statement-II are false.

4) Value of Λ_m^∞ for $SrCl_2$ in water at 25°C from the following data :

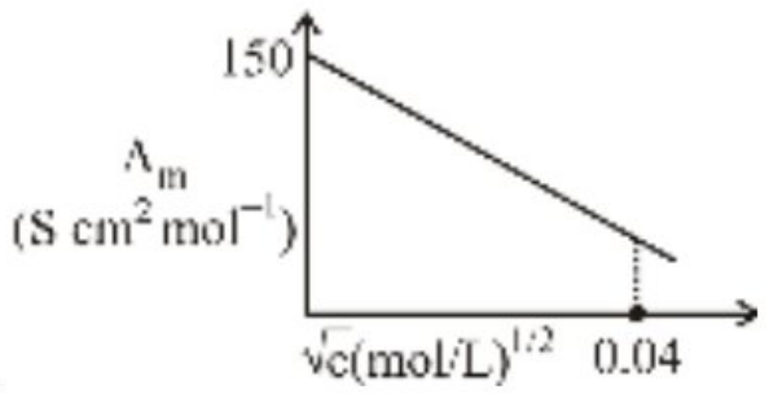
Conc. (mol/L)	0.25	1
Λ_m (Ω^{-1} cm^2 mol^{-1})	260	250

(1) 270

(2) 260

(3) 250

(4) 255



5) At 298 K, graph between Λ_m and $\sqrt{c}$ for KCl is given. Limiting molar conductance for KCl will be-

(1) 600 S cm^2 mol^{-1}

(2) 150 S cm^2 mol^{-1}

(3) 300 S cm^2 mol^{-1}

(4) 0.04 S cm^2 mol^{-1}

6) Λ°_m of $BaCl_2$, $Ba(OH)_2$ and NH_4Cl are x_1, x_2 and x_3 respectively. Find Λ°_m of NH_4OH

(1) $(x_3 + x_2 - x_1)$

(2) $(2x_3 + x_2 - x_1)$

(3) $\frac{(2x_3 + x_2 - x_1)}{2}$

(4) $\frac{(2x_2 + x_3 - x_1)}{2}$

7) The conductivity of a saturated solution of $BaSO_4$ is 3.06×10^{-6} ohm^{-1} cm^{-1} and its equivalent conductance is 1.53 $ohm^{-1}cm^2eq^{-1}$. The k_{sp} for $BaSO_4$ will be :-

(1) 4×10^{-12}

(2) 2.5×10^{-13}

(3) 25×10^{-9}

(4) 10^{-6}

8) Find the standard potential of cell at 298 K

$Cu^{+2}(aq) + In^{+2}(aq) \Rightarrow Cu^{+}(aq) + In^{+3}(aq)$

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Given that $E^{o}_{Cu^{+2}/Cu^{+}} = 0.15\,V$, $E^{o}_{In^{+3}/In^{+}} = -0.42\,V$, $E^{o}_{In^{+2}/In^{+}} = -0.40\,V$

(1) 0.55 V

(2) 0.57 V

(3) 0.59 V

(4) 0.95 V

9)

When aqueous solution of NaC□ is electrolysed

(1) resultant solution has pH > 7

(2) resultant solution turns red litmus blue

(3) H_2 and C□$_2$ are the products

(4) all the given choices are correct

10) Review the SRP (at 25°C) data in acidic medium

$Ti^{4+} + e \rightarrow Ti^{3+}, E^{\circ} = -xV$,
$Fe^{3+} + e \rightarrow Fe^{2+}, E^{\circ} = -yV$, } where $x < y$

point out the wrong statement :-

(1) Ti^{3+} will act as reductant

(2) Fe^{3+} will act as oxidant

(3) $[Ti^{3+} + Fe^{3+}]$ is a spontaneous process

(4) Ti^{4+} will act as oxidant

11) Other things being equal, the life of a Daniell cell may be increased by :–

(1) Keeping low temperature

(2) Using large copper electrode

(3) Decreasing concentration of copper ions

(4) Using large zinc electrodes

12) How many statements are correct for Daniel cell?
(i) In external circuit electric current flow from cathode to anode.
(ii) Anode is negatively polarized.
(iii) Mobility of cation and anion present in salt bridge is nearly same
(iv) Concentration of cation in anodic compartment decreases.
(v) Concentration of cation in cathodic compartment decreases.

(1) 2

(2) 3

(3) 4

(4) 5

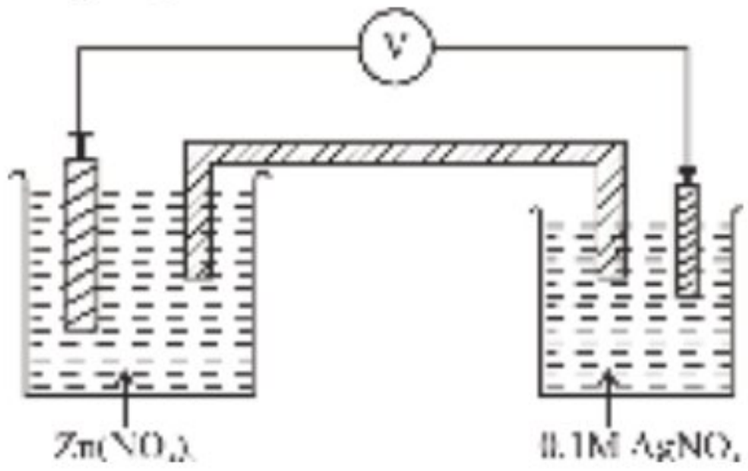
13) If following cell is completely discharge, what is the concentration of Zn^{+2} :-

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$E^0_{Ag^+/Ag} = 0.8\,V, E^0_{Zn^{+2}/Zn} = -0.7\,V$



(1) 10^{31}

(2) 10^{40}

(3) 10^{48}

(4) 10^{56}

14)

For cell

$Pt_1\ H_{2(g)}\ (P_1 atm)\ H^+(C_1)||H^+(C_2)|H_2(P_2 atm),Pt$

For spontaneous Process condition should be If $P_1 = P_2 = 1$ atm ?

(1) $C_1 > C_2$

(2) $C_1 = C_2$

(3) $C_2 > C_1$

(4) Can't predict

15) The standard electrode potential for the following reaction is +1.33V. What is the potential at pH = 2.0

$Cr_2O_7^{2-}$ + (aq 1M) + $14H^+$ (aq) + $6e^-$ →

$2Cr^{+3}$ (aq,1M) + $7H_2O$ (□)

(1) 1.820 V

(2) + 1.990 V

(3) + 1.608 V

(4) + 1.0542 V

16) Given that :

Ni^{2+} | Ni = –0.25 V;

Cu^{2+} | Cu = 0.34 V

Ag^+ | Ag = 0.80 V;

Zn^{2+} | Zn = – 0.76 V

Which of the following reactions under standard condition will not take place in the specified direction :

(1) $Ni^{2+}_{(aq.)} + Cu_{(s)} \rightarrow Ni_{(s)} + Cu^{2+}_{(aq.)}$

(2) $2Ag^{+}_{(aq.)} + Cu_{(s)} \rightarrow 2Ag_{(s)} + Cu^{+2}_{(aq.)}$

(3) $Zn_{(s)} + Cu^{+2}_{(aq.)} \rightarrow Zn^{+2}_{(aq.)} + Cu_{(s)}$

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(4) $2H^+_{(aq.)} + Zn_{(s)} \rightarrow H_{2(g)} + Zn^{2+}_{(aq.)}$

17) The SRP for $Zn^{2+}|Zn$, $Ni^{2+}|Ni$ and $Fe^{2+}|Fe$ are –0.76 V, –0.23 V and –0.44 V respectively. The reaction $X + Y^{2+} \rightarrow X^{2+} + Y$ will spontaneous when :

(1) X = Ni, Y = Fe

(2) X = Ni, Y = Zn

(3) X = Fe, Y = Zn

(4) X = Zn, Y = Ni

18) Efficiency of the following cells is 84%

$A(s) + B^{2+}(aq) \rightleftharpoons A^{2+}(aq) + B(s)$, $\Delta H = -285$ kJ

Then the standard electrode potential of the cell will be :-

(1) 1.20 V

(2) 2.40 V

(3) 1.10 V

(4) 1.24 V

19) Given the data at 25°C

$Ag + I^- \rightarrow AgI + e^-$; $E° = 0.152$ V

$Ag \rightarrow Ag^+ + e^-$; $E° = -0.800$ V

What is the value of log K_{sp} for AgI ?

(1) –8.12

(2) –37.82

(3) +8.612

(4) –16.13

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20) How many litres of chlorine gas will be obtained by electrolysis of molten NaCl at STP if the electrolysis continued for 9.65 sec using 1000 amp current :

(1) 11.2 L

(2) 1.12 L

(3) 1800 L

(4) 1231.6 L

21) An electric current is passed through silver nitrate solution using silver electrodes. 15.28 g of silver was found to be deposited on cathode. What will be the weight of copper deposited on cathode if same amount of electricity is passed through copper sulphate solution using copper electrodes ?

(1) 4.49 g

(2) 6.4 g

(3) 12.8 g

(4) 3.2 g

22) Cost of electricity for the production of X litre H_2 at STP at cathode is Rs. X, then cost of

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electricity for the production of X litre O_2 gas at STP at anode will be (assume 1 mole of electrons as one unit of electricity)

(1) 2 x

(2) 4 x

(3) 16 x

(4) 32 x

23) Which of the following property is CORRECT for $edta^{-4}$ ligand :-

(1) Anionic ligand

(2) Hexadentate ligand

(3) It forms 5 rings

(4) All are correct

24) In the isoelectronic series of metal carbonyl, the CO bond strength is expected to increase in the order:

(1) $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^-$

(2) $[V(CO)_6]^- < [Cr(CO)_6] < [Mn(CO)_6]^+$

(3) $[V(CO)_6]^- < [Mn(CO)_6]^+ < [Cr(CO)_6]$

(4) $[Cr(CO)_6] < [Mn(CO)_6]^+ < [V(CO)_6]^-$

25) A compound whose IUPAC name is not correctly written is :-

	Complex	**Name**
(1)	$Fe\left(\eta^5 - C_5H_5\right)_2$	Bis(η^5-cyclopentadienyl) iron(0)
(2)	$Cr(C_6H_6)_2$	Bis(η^6-benzene) chromium(0)
(3)	$[CoCl_2(H_2O)_4]^+$	Tetraaquadichlorocobalt (III) ion
(4)	$[Zn(NCS)_4]^{2-}$	Tetrathiocyanato-N-zincate (II) ion

(1) 1

(2) 2

(3) 3

(4) 4

26) Hybridisation, magnetic nature & magnetic moment (spin–only) for $[Co(C_2O_4)_3]^{-3}$ is ?

(1) sp^3d^2 ; paramagnetic ; 4.8 B.M.

(2) sp^3d^2 ; diamagnetic ; 0 B.M.

(3) d^2sp^3 ; paramagnetic ; 4.8 B.M.

(4) d^2sp^3 ; diamagnetic ; 0 B.M.

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27) Carbon donor ligands are strong ligands and usually forms low spin complexes. Among the complexes given below, select the high spin complex :-

(1) $Fe(CN)_6]^{3-}$

(2) $[Fe(CN)_6]^{4-}$

(3) $[Fe(C_2O_4)_3]^{3-}$

(4) none of these

28) (Valence Bond Theory) The coordination complex $[Cu(OH_2)_6]^{2+}$ has one unpaired electron. Which of the following statements are true ?

(1) The complex is octahedral
(2) The complex is an outer orbital complex
(3) The complex is d^2sp^3 hybridized
(4) The complex is diamagnetic
(5) The coordination number is 6

(1) 1, 4

(2) 1, 2, 5

(3) 2, 3, 5

(4) 2, 3

29) The spin only magnetic moment of $[MnBr_4]^{2-}$ is 5.9 B.M. The correct statement about the complex is –

(1) It has tetrahedral gemoetry

(2) Configuration is $t_2g^5 eg^0$

(3) Configuration is $t_2g^2 eg^3$

(4) It is low spin complex

30) If the crystal field stabilisation energy (CFSE) for $[FeCU]^{2-}$ is x J, then the CFSE for $[FeCb]^{4-}$ will be

(1) $\frac{4x}{9}J$

(2) $\frac{9x}{4}J$

(3) $\frac{2x}{3}J$

(4) $\frac{9x}{5}J$

31) When 0.1 mol $CoCl_3(NH_3)_5$ is treated with excess of $AgNO_3$, 0.2 mol of AgCl are obtained. The conductivity of solution will correspond to :

(1) 1 : 3 electrolyte

(2) 1 : 2 electrolyte

(3) 1 : 1 electrolyte

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(4) 3 : 1 electrolyte

32) The structure of $[Cu(H_2O)_4]^{+2}$ ion is -

(1) Tetrahedral
(2) Square planar
(3) Octahedral
(4) Pyramidal

33) Which of the following are the Limitations of VBT for coordination compounds?
I. It does not give quantitative interpretation of magnetic data.
II. It does not distinguish between weak and strong ligands.
III. It does not explain the colour exhibited by coordination compounds.
IV. It does not give a quantitative interpretation of the thermodynamic or kinetic stabilities of coordination compounds.
Identify correct option from given choices :

(1) I, II, III and IV
(2) II, III and IV
(3) I, II and IV
(4) II, III and IV

34) Assertion (A): VBT can explain colour of coordination compounds.
Reason (R): Colour arises due to d–d transitions.

(1) Both A and R true, R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

35) Which of the following is correct for Zeise's salt:-

(1) The complex ion has square planar geometry
(2) Pt is in + 2 oxidation state
(3) $H_2C = CH_2$ molecule is perpendicular to $PtCl_3$ plane
(4) All of the above

36) Which of following is not considered as organometallic compound ?

(1) Ferrocene
(2) Ziese's salt
(3) Grignard reagent
(4) Cis-platin

37) Metal-carbon bond length is maximum in which of the given organometallic species?

(1) $[Mn(CO)_6]^+$

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NOVA

(2) $[V(CO)_6]^-$

(3) $[Ti(CO)_6]^{2-}$

(4) $[Cr(CO)_6]$

38)

Which of the following is correct statement regarding biological importance of coordination compounds ?

(1) Chlorophyll is a coordination compound of Fe

(2) Vitamin B_{12} , cyanocobalamine is a coordination compound of Mg

(3) Haemogolobin is a coordination compound of Co

(4) Carboxypeptidase is a coordination compound of Zn

39) A blue colouration is obtained when :-

(1) NH_4OH is added to $CuSO_4$

(2) $FeCl_3$ reacts with $K_4[Fe(CN)_6]$

(3) Anhydrous white $CuSO_4$ is dissolved in water

(4) All of these

40) Dimethyl glyoxime gives a red precipitate with (Ni^{+2}) ion which is used for its detection. To get this precipitate readily, the best pH range is :-

(1) < 1

(2) 2 – 3

(3) 3 – 4

(4) 9 – 11

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41) ***Assertion*** :- Coordination number of Pt in Zeise's salt is 4. ***Reason*** :- C_2H_4 act as bidentate ligand

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) **Assertion** is false but **Reason** is true.

42) The solubility of AgBr in hypo solution is due to the formation of :-

(1) Ag_2SO_3

(2) $Ag_2S_2O_3$

(3) $[Ag(S_2O_3)]^-$

(4) $[Ag(S_2O_3)_2]^{3-}$

43) The complex [$Fe(H_2O)_5$ NO]$^{2+}$ is formed in the 'brown ring test' for nitrates. Choose the incorrect statement for the complex.

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(1) Its magnetic moment is approximately 3.9 B.M.

(2) The oxidation state of iron is + 1

(3) The hybridisation of central metal ion is $sp^3 d^2$

(4) The hybridisation of central metal Ion is d^2sp^3

44) Which among the following statement is wrong for complex Prussian blue $Fe_4[Fe(CN)_6]_3$

(1) Magnetic moment of counter ion of Fe is 5.9 BM

(2) It has coordination number 6

(3) Magnetic moment of Fe in complex ion is 1.73 BM

(4) It is homoleptic complex

45) Which complex has maximum spin?

(1) $[CoF_6]^{3-}$

(2) $[Ni(CN)_4]^{-4}$

(3) $[Co(ox)_3]^{-3}$

(4) $[Co(NH_3)_6]^{+3}$

BIOLOGY

1) Consider the following four statements (a-d) and select the option which includes all the **incorrect** ones only.
(a) Each testicular lobule contains 750 highly coiled seminiferous tubules, in which sperms are produced.
(b) Sertoli cells undergo meiotic divisions finally leading to sperm formation.
(c) The regions outside the seminiferous tubules called interstitial spaces, contain small blood vessels, Leydig cells, and some immunologically competent cells.
(d) The scrotum helps in maintaining the low temperature of the testes, necessary for spermatogenesis.

(1) Statements (b), (c) and (d).

(2) Statements (a), (b).

(3) Statements (a), (b) and (c).

(4) Statements (a), (c) and (d).

2) **Assertion :** Absence of menstruation is not a confirmatory indicator of pregnancy. **Reason :** Absence of menstruation can happen due to poor health or stress.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

3) Identify the correct match from Column I, II and III.

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	Column I		Column II		Column III
(A)	Ovulatory phase	(a)	5th-13th day	(i)	Breakdown of endomentium
(B)	Menstruation phase	(b)	15th-28th day	(ii)	Release of secondary oocyte
(C)	Secretory phase	(c)	14th day	(iii)	Development of mature follicle
(D)	Follicular phase	(d)	1st-4th day	(iv)	Formation of corpus luteum

(1) A-c-i, B-d-ii, C-a-iii, D-b-iv

(2) A-c-ii, B-a-i, C-b-iii, D-d-iv

(3) A-d-i, B-a-iv, C-d-iv, D-c-ii

(4) A-c-ii, B-d-i, C-b-iv, D-a-iii

4) Find out the correct match from the following table :

Column-I	Column-II	Column-III
(i) Intestitial cells	Cells of seminiferous tubule	Secrete testosterone
(ii) Nurse cells	Cells of testis	Provide nutrition to germ cells
(iii) Spermatocyte	Cells of seminiferous tubule	Divided by meiosis and produce sperms
(iv) Cremaster muscles	Scrotum	Help in elevation of testis

(1) Only (i) and (iii)

(2) Only (ii) and (iii)

(3) (ii), (iii) and (iv)

(4) (i), (ii), (iii) and (iv)

@examly

5) Which is/are functions of testosterone hormone?
(a) Controls growth and development
(b) Stimulates bone density
(c) Protein catabolism
(d) Sperm maturation
(e) Development of secondary sexual characters

(1) c, d and e only

(2) a, b and e only

(3) a, c and e only

(4) a, b, d and e only

6) Read the following four statements (A-D):-
(A) Process of formation of mature female gamete is called oogenesis
(B) At puberty 60,000 to 80,000 total primary follicles are left in both ovaries
(C) No oogonia are formed or added after birth in female
(D) Process of gametogenesis begins in female at puberty
Which of the above statements are **correct** ?

(1) A and C

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(2) B and D

(3) A, C and D

(4) B, C and D

7) Arrange following events of female reproductive cycle in (human) in sequence :-
(I) Secretion of FSH
(II) Growth of corpus luteum
(III) Growth of follicles
(IV) Ovulation
(V) Sudden increase in level of LH.

(1) III → I → IV → II → V

(2) I → III → V → IV → II

(3) I → IV → III → V → II

(4) II → I → III → IV → V

8) Identify the correct match from the column I, II and III :-

	Column-I		Column-II		Column-III
1	Labia majora	(a)	Thin fold like structure of skin	(i)	homologous to penile urethra
2	Labia minora	(b)	Small, Cylindrical and erectile	(ii)	Homologous to scrotum
3	Clitoris	(c)	Folds of subcutaneous tissue	(iii)	Homologous to cowper's gland
4	Bartholin's gland	(d)	Greater vestibular gland	(iv)	Homologous to penis

(1) 1-c-ii, 2-a-i, 3-b-iv, 4-d-iii

(2) 1-d-iv, 2-c-ii, 3-a-i, 4-b-iii

(3) 1-a-i, 2-b-ii, 3-c-iii, 4-d-iv

(4) 1-b-i, 2-a-iii, 3-d-iv, 4-c-iii

9) Read the following statements carefully.
The .....(A)..... present on the inside wall of .....(B)..... multiply by .....(C)..... division. A .....(D)..... completes the .....(E)..... division leading to formation of two equal, haploid cells called .....(F)..... .
Which one of the following options gives the correct filling of A, B, C, D, E and F?

	(1)	(2)	(3)	(4)
A	Spermatogonia	Primary Spermatocyte	Spermatogonia	Spermatogonia
B	Seminiferoustubule	Sertoli Cell	Sertoli Cell	Seminiferous tubule
C	Mitotic	Mitotic	First meiotic	First meiotic
D	Primary Spermatocyte	Secondary Spermatocyte	Secondary Spermatocyte	Spermatid
E	First meiotic	First meiotic	Second meiotic	Second meiotic
F	Secondary Spermatocyte	Spermatid	Spermatid	Second meiotic Sper

(1) 1

(2) 2

(3) 3

(4) 4

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10) Read the given statements (A-D) carefully:
(A) Hymen is the reliable indicator of virginity.
(B) Acrosome of sperm contains several mitochondria.
(C) After fertilization, corpus luteum transforms into Graafian follicle.
(D) The process of delivery of foetus is called as gestation.
How many statements are incorrect among these?

(1) 1
(2) 2
(3) 3
(4) 4

11) **Assertion (A):** Corpus luteum start to degenerate if the ovum remain unfertilized.
Reason (R): Gonadotropins hormones maintain the corpus luteum upto 22nd day of menstrual cycle.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
(2) **(A)** is correct but **(R)** is not correct
(3) **(A)** is incorrect but **(R)** is correct
(4) Both **(A)** and **(R)** are correct but **(R)** is the correct explanation of **(A)**

12) The process of childbirth is called parturition which is induced by a complex neuroendocrine mechanism involving these hormones except

(1) Estrogen
(2) Oxytocin
(3) CRH
(4) hPL

@examlix

13) **Statement - I :** The male accessory glands include paired seminal vesicles, a prostate and paired bulbourethral glands.
Statement - II : The secretions of bulbourethral glands also helps in the lubrication of the penis.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement-I and statement-II are incorrect.
(2) Statement-I is incorrect but statement-II is correct.
(3) Statement-I is correct but statement-II is incorrect.
(4) Both Statement-I and statement-II are correct.

14) A female of 24 years age attained puberty at the age of 14 years. She has two children born by normal delivery. How many polar body are formed in her body till date (each menstrual cycle is of 30 days) :-

(1) 80
(2) 100
(3) 92
(4) 70

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15) Which one of the following is the correct matching of the events occuring during menstrual cycle ?

(1) Proliferative phase – Rapid regeneration of myometrium and maturation of Graafian follicle

(2) Luteal phase – Development of corpus luteum and increased secretion of progesterone

(3) Menstruation – Breakdown of perimetrium and ovum fertilized

(4) Ovulation – LH and progesterone attain peak level

16) In human female the blastocyst :
(A) Form placenta after implantation.
(B) Gets implanted into uterus three days after ovulation.
(C) Gets nutrition from uterine endometrial secretions only after implantation.
(D) Gets implanted in endometrium by trophoblast.
Chosse an correct set of answer

(1) a and b

(2) a and c

(3) a and d

(4) c and d

17) Read the following statements (i to iv) and answer the following question.
(i) Each testes has highly coiled 250 compartments called seminiferous tubules.
(ii) Erection of the penis due to presence of special tissues facilitates insemination
(iii) Immunologically competent cells are also present in the interstitial spaces of seminiferous tubules.
(iv) Testes lie outside the abdominal cavity in a pouch like skin called scrotum.
(v) Bulbourethral gland is a single accessory gland.
How many of the above statements are incorrect?

(1) (i), (ii) and (iii)

(2) (iii) and (v)

(3) (i) and (v)

(4) (ii), (iv) and (v)

18) Read the following statements (i to v) and answer the following question.
(i) The structure is also called womb.
(ii) Its shape is like an inverted pear.
(iii) The process of fertilization takes place in this structure.
(iv) The wall of this structure has three layers of tissue.
(v) It secretes several steroid hormones.
Identify the correct characteristics feature regarding uterus from the above statements.

(1) (i) and (iv)

(2) (iii) and (v)

(3) (i), (ii) and (iv)

(4) All the five statements

19) Match the columns I and II and choose the CORRECT combination from the option given.

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	Column I		Column II
a	Testis	i	Erectile
b	Infundibulum	ii	Oval
c	Uterus	iii	Finger-like
d	Fimbriae	iv	Funnel-shaped
e	Clitoris	v	Inverted pear like

(1) a (ii), b (v), c(i), d(iii), e(iv)

(2) a(i), b(iv), c(v), d(ii), e(iii)

(3) a(ii), b(iv), c(v), d(iii), e(i)

(4) a(i), b(iv), c(v), d(iii), e(ii)

20) How many of the following statements is/are not correct?
(A) Presence or absence of hymen is a reliable indicator of virginity/sexual experience
(B) Vaginal opening is compeletly closed by a septum of mucus membrane called hymen
(C) The uterus open into fibromuscular and glandular tube like organ called vagina
(D) Vagina and cervical canal form a birth canal

(1) 4

(2) 3

(3) 2

(4) 1

21) Select correct option regarding Oogenesis :-

(1) Oogenesis is initiated after birth of the female child

(2) The gamete mother cells (Oogonia), starts division and enter into prophase-I of meiotic division, gets temporarily arrested at this stage called primary oocyte

(3) At birth 6 to 8 millions primary follicles are present in each ovary

(4) Secondary follicle is characterised by fluid filled cavity called Antrum

22)

Arrange the correct sequence of fertilisation process :-
(A) Intermixing of chromosomes
(B) Fusion of pronuclei of sperm & ovum
(C) Intermixing of cytoplasm

(D) Union of male & female gametes
(1) D→B→C→A

(2) C→D→B→A

(3) C→B→A→D

(4) D→C→B→A

23)

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Which one of the following is the correct matching of the events occuring during menstrual cycle ?

(1)	Menstruation	Breakdown of myometrium and ovum not fertilised
(2)	Ovulation	LH and FSH attain peak level and sharp fall in the secretion of progesterone.
(3)	Proliferative phase	Rapid regeneration of myometrium and maturation of Graafian follicle.
(4)	Development of corpus luteum	Secretory phase and increased secretion of progesterone

(1) 1

(2) 2

(3) 3

(4) 4

24) Match the following and choose the correct answer :

(A)	Fusion of male & female gamete	(i)	Fundus of uterus
(B)	Attachment of blastocyst	(ii)	Ovary
(C)	Reactivation of sperms	(iii)	Ampulla
(D)	Development of Graafian follicle	(iv)	Vagina

(1) A-(ii), B-(iii), C-(i), D-(iv)

(2) A-(ii), B-(i), C-(iv), D-(iii)

(3) A-(iii), B-(ii), C-(iv), D-(i)

(4) A-(iii), B-(i), C-(iv), D-(ii)

25)

Identify the correct match from the columns I, II and III.

	Column-I		**Column-II**		**Column-III**

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A	Leydig cells	a	Ovary	i	Androgen binding protein
B	Sertoli cells	b	Semini-ferous tubule	ii	Progesterone
C	Theca interna	c	Testis	iii	Androgens
D	Corpus luteum	d	Follicle	iv	Estrogen

(1) A → d → i; B → c → iv; C → b → iii; D → a → ii

(2) A → c → iii; B → b → iv; C → d → ii; D → a → i

(3) A → b → i; B → c → iii; C → a → iv; D → d → ii

(4) A → c → iii; B → b → i; C → d → iv; D → a → ii

26) Read the following statements carefully and answer the question given below.
(A) Estrogen increase the protein anabolism and its moderate level inhibit the FSH and LH.
(B) progesterone secreted by corpus luteum and its moderate level inhibit the secretion of GnRh.
(C) Inhibin is sereted by corpus luteum and it inhibit the secretion of FSH and to Lesser extent LH.
(D) Relaxin release by corpus luteum and its moderate level inhibit the secretion of GnRh, FSH, LH.
How many statement are correct.

(1) A & B

(2) C & D

(3) A, B & C

(4) B, C & D

27) Mark the incorrect statement :-
(A) Sertoli cells provide nourishment to dead sperm
(B) Vagina secrete alkaline fluid, to make easy passage for sperms
(C) Fertilisation occurs in fundus of female uterus
(D) Hymen is the reliable indicator of virginity

(1) A,C

(2) B,C,D

(3) A,B,C and D

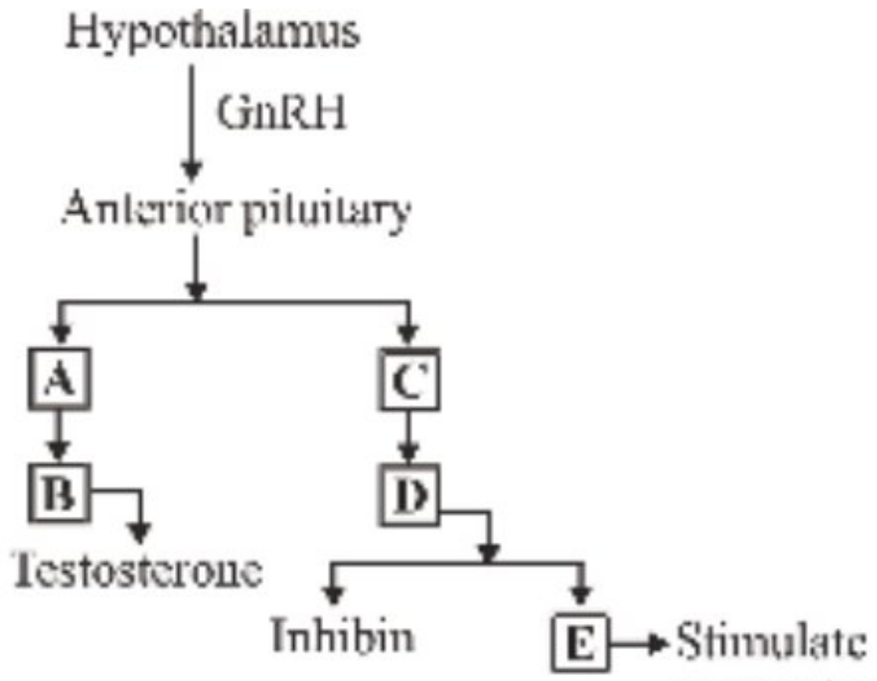
(4) A,B

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28) Find out **A, B, C, D** and **E** respectively :-

	A	B	C	D	E
(1)	Testis	Leyding cells	Sertoli cells	AMH	ABP
(2)	ICSH	Testis	AMH	Sertoli cells	ABP
(3)	LH	Interstitial cells	FSH	Sertoli cells	AMH
(4)	ICSH	Leyding cells	FSH	Sertoli cells	ABP

(1) 1

(2) 2

(3) 3

(4) 4

29)

Correctly matched pairs are :-
(a) Clitoris – Erectile body in female and homologous to penis of male
(b) Colostrum – Secretion of preputial gland
(c) Oxytocin – Milk synthesis hormone

(d) Labia majora – Homologous to scrotum

(1) a and d

(2) a and c

(3) a and b

(4) c and d

30) Choose the incorrect option w.r.t amniocentesis.

(1) Statutory banned in India or sex determination

(2) Pre-natal diagnostic test

(3) Can determine the survivability of the foetus

(4) Can be usually performed during any week of gaestation

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31) Which of the following statements releated to fertilization, implantation, and hormonal regulation is incorrect ?
A. Fertilization occurs only when sperm and ovum meet at the ampullary-isthmic junction of the cervix
B. hCG maintains the corpus luteum and stimulates estrongen and progesterone secretion.
C. High levels of estrogen and progesterone prevent menstruation and support endometrial thickening.
D. Low levels of FSH and LH during the luteal phase prevent development of new follicles.

(1) A only
(2) A and B only
(3) D only
(4) B and C only

32)

In humans, _____ (A) does not store excretory wastes, unlike in _____ (B), while _____ (C) keeps the foetus suspended in fluid and _____ (D) helps in early blood cell formation.

(1) A–Allantois, B–Reptiles and birds, C–Amnion, D–Yolk sac
(2) A–Yolk sac, B–Mammals, C–Chorion, D–Allantois
(3) A–Amnion, B–Humans, C–Allantois, D–Chorion
(4) A–Chorion, B–Amphibians, C–Yolk sac, D–Amnion

33)

Fill in the blanks and choose the correct option:

After implantation, _____ (A) gives rise to chorionic villi that embed into _____ (B), while _____ (C) forms blood vessels connecting embryo with placenta and _____ (D) reduces after 8 weeks.

(1) A–Amnion, B–Placenta, C–Yolk sac, D–Chorion
(2) A–Chorion, B–Uterine endometrium, C–Allantois, D–Yolk sac
(3) A–Allantois, B–Endoderm, C–Chorion, D–Amnion
(4) A–Yolk sac, B–Uterus, C–Amnion, D–Allantois

34)

Fill in the blanks and choose the correct option:

The _____ (A) secretes amniotic fluid which prevents _____ (B), while _____ (C) forms chorionic villi and _____ (D) acts as the first haematopoietic organ.

(1) A–Amnion, B–Desiccation, C–Chorion, D–Yolk sac

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(2) A–Chorion, B–Hydration, C–Amnion, D–Allantois

(3) A–Yolk sac, B–Desiccation, C–Allantois, D–Chorion

(4) A–Allantois, B–Diffusion, C–Yolk sac, D–Amnion

35) Study the following statements and answer the question. In a process called 'A', 'B' division starts as the zygote moves through the 'C' of the 'D' towards the 'E'.
Identify A, B, C, D and E.

(1) A→Blastulation, B→Meiotic, C→Ampulla, D→Fallopian tube, E→Uterus

(2) A→Parturition, B→Meiotic, C→Infundibulum, D→Uterus, E→Vagina

(3) A→Implantation, B→Mitotic, C→Fimbriae, D→Ovary, E→Cervix

(4) A→Cleavage, B→Mitotic, C→Isthmus, D→Oviduct, E→Uterus

36) Which one of the following statements is correct?

(1) Fertilisation in human takes place in the womb

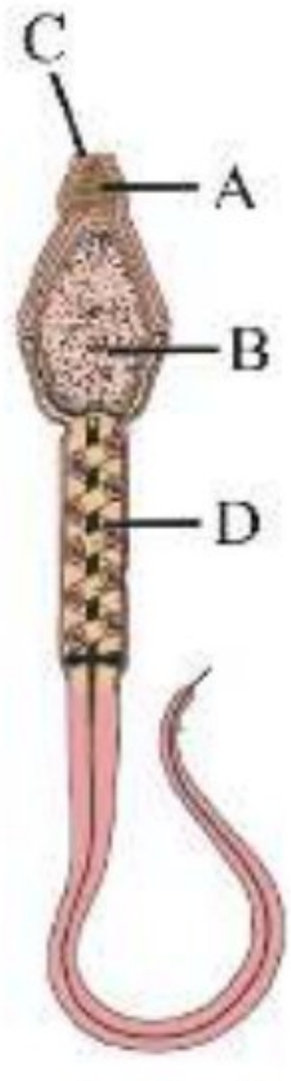
(2) Zygote contains haploid number of chromosomes

(3) Changes in zona pellucida prevent polyspermy

(4) Primary oocyte inhibits the process of Oogenesis

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37)



Recognise the figure and find out the correct matching

(1) C-Plasma membrane, B-Acrosome, A-Nucleus, D-Mitochondria

(2) C-Plasma membrane, A-Acrosome, B-Nucleus, D-Mitochondria

(3) A-Plasma membrane, B-Acrosome, D-Nucleus, C-Mitochondria

(4) D-Plasma membrane, A-Nucleus, B-Acrosome, C-Mitochondria

38) Read the following statements A-D.
(A) Vigorous contraction of the uterus at the end of the pregnancy cause delivery of the foetus.
(B) Parturition is induced by a complex neuroendocrine mechanism.
(C) The signals for parturition originated only from fully developed foetus.

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(D) Breast-Feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby.
How many of the above statements are correct ?

(1) Two

(2) Four

(3) One

(4) Three

39) Which of the following statements are not correct?
(A) Leydig cells synthesise & secrete testicular hormones called androgens.
(B) Sertoli cells provide nutrition to the interstitial cells.
(C) Each fallopian tube is about 10-12 cm long and extends from periphery of each ovary to the uterus.
(D) Urethra is a part of female reproductive tract.
(E) The human male ejaculates about 200 to 300 billion sperms during a coitus.

(1) A, C, E

(2) B, C, D

(3) A, C

(4) B, D, E

40) Which of the following event doesn't occur in case of pregnancy ?
(A) Decrease in level of progesterone.
(B) Degeneration of corpus luteum.
(C) Stoppage of menstrual cycle.
(D) Endometrial hyperplasia and Myometrium atrophy.
(E) Increase in level of thyroxine in maternal blood.

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(1) A, B and C

(2) B, D and E

(3) A, B and D

(4) A, B and E

41) **Assertion :** Ovarian follicles in different stages of development are embedded in the stroma
Reason : The female external genitalia includes mons pubis, labia majora, labia minora, hymen and clitoris

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false.

42) Read the following statements and select the correct option :
Statement A: Humans are sexually reproducing and viviparous organisms.
Statement B: Cyclic menstruation is an indicator of normal reproductive phase and extends between menarche and menopause.

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(1) Both statements A and B are correct

(2) Both.statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement-B is correct

43) **Assertion (A)**: In humans, testes are situated outside the abdominal cavity within the scrotum.
Reason (R): The scrotum helps in maintaining the low temperature of the testes necessary for spermatogenesis.
In the light of above statements, select the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true, (R) is false

(4) Both (A) and (R) are false

44) Select the incorrect statement:

(1) Abortions can happen spontaneously too.

(2) Infertility is defined as the ability to produce an inviable offspring and is always due to defects in male partner.

(3) Creating awareness about sex related aspects is an effective method to improve reproductive health of the people.

(4) Continued intense lactation could help as a natural method of contraception upto maximum six months from parturition.

45) Read the following statements regarding the scrotum in humans:
Statement A: The scrotum is a pouch situated outside the abdominal cavity that houses the testes.
Statement B: The low temperature maintained by the scrotum is essential for the process of spermiation and the secretion of testosterone. Which of the following is correct?

(1) Both Statement A and Statement B are correct.

(2) Statement A is correct, but Statement B is incorrect.

(3) Statement A is incorrect, but Statement B is correct.

(4) Both Statement A and Statement B are incorrect.

46) Sahiwal cow is result of :

(1) Natural selection of wild cow

(2) Artificial selection of wild cow

(3) Somatic variation

(4) Test cross

47) A true-breeding line or pure line is one that has undergone continuous_____pollination and show the_____trait for several generations.

(1) Self, stable

(2) Fluctuating, stable

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(3) Cross, fluctuating

(4) Self, fluctuating

48) Mendel's work remain unrecognised till 1900 due to all except :-

(1) His work could not be widely publicised

(2) His work was not supported with required data

(3) He could not provide any proof for existence of factors

(4) His concept of genes as stable and discrete units was not accepted by his contemporaries

49) Dominance of any character generally depends on :-

(1) Gene or product related informations of any gene

(2) Character choosen by ourself in study

(3) Environmental factors

(4) Both 1 and 2

50) The fact that the alleles do not show any blending and that both the characters are recovered as such in F_2 generation, become the basis of :-

(1) Law of Dominance

(2) Law of paired factors

(3) Law of segregation

(4) Law of independent assortment

51) Graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross, is known as :-

(1) Mendel square

(2) Punnett square

(3) Crossboard method

(4) Emasculation method

52) 25 : 46 : 26 genotypic ratio in F_2 generation is expected from which parental cross?

(1) Aa × Aa

(2) AA × Aa

(3) AA × aa

(4) AA × AA

53) Which of the following seed characters were studies by Mendel in his experiments ?
(a) Seed shape
(b) Seed weight
(c) Seed coat colour
(d) Seed starch grain size
(e) Seed number

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(1) a & b

(2) a & e

(3) a & c

(4) only a

54) Distance between the genes a, b, c and d in map units is a-d = 3.5, b-c = 1, a-b = 6, c-d = 1.5 and a-c = 5. Find out the sequence of the genes.

(1) adcb

(2) acdb

(3) abed

(4) acbd

55) What is the probability of obtaining pure homozygous individuals, if a cross is made between AaBbCcDd × aaBBCCdd individuals ?

(1) $\frac{2}{16}$

(2) $\frac{1}{16}$

(3) $\frac{16}{248}$

(4) $\frac{1}{64}$

56)

A dihybrid test cross produces following results in three different conditions A, B and C, then identify A, B and C :-

GgTt × ggtt

↓

A - 1 : 1 : 1 : 1

B - 9 : 1 : 1 : 9

C - 1 : 1

(1) A - Complete linkage
B - Independent assort C - Incomplete linkage

(2) A - Independent assortment
B - Complete linkage C - Incomplete linkage

(3) A - Independent assortment
B - Incomplete linkage C - Complete linkage

(4) A - Incomplete linkage
B - Complete linkage C - Independent assortment

57) In Drosophila melanogaster, vestigial wings are determined by a recessive allele of a gene that is linked to a gene with a recessive allele that determines black body color. T. H. Morgan crossed black-bodied, normal-winged females and gray-bodied, vestigial-winged males. The F_1 were all gray bodied, normal winged. The F_1 females were crossed to homozygous recessive males to produce testcross progeny. Morgan calculated the map distance to be 17 map units. Which of the following

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information is correct about the testcross progeny?

(1) black-bodied, normal-winged flies = 17% of the total

(2) black-bodied, normal-winged flies plus gray-bodied, vestigial-winged flies = 17% of the total

(3) gray-bodied, normal-winged flies plus black-bodied, vestigial-winged flies = 17% of the total

(4) black-bodied, vestigial-winged flies = 17% of the total

58) What phenotypic ratio is obtained by selfing of a dihybrid, in which one genes is completely dominant and one gene is incompletely dominant ?

(1) 18 : 6 : 6 : 2 : 9 : 3 : 3 : 1 : 9 : 3 : 3 : 1

(2) 27 : 9 : 9 : 9 : 3 : 1 : 3 : 3

(3) 3 : 6 : 3 : 1 : 2 : 1

(4) 18 : 6 : 6 : 2 : 9 : 3 : 3 : 1 : 1 : 2 : 1

59) Which of the following statements is **true** regarding the inheritance pattern of **Colour Blindness**?

(1) It occurs in about 8% of females and 0.4% of males.

(2) A daughter will be colour blind only if her mother is colour blind and father is normal.

(3) A carrier mother can pass the gene to her sons, who have a 50% chance of being colour blind.

(4) It is an autosomal dominant disorder.

60) When parents P_1 and P_2 were crossed, F_2 progeny was produced with three fourth similar features in phenotype to P_2 and F_1 and one fourth possessed contrasting traits. If the traits being considered here are for height with 'T' for tall and 't' for short. What wil be the possible genotype of P_1 and one fourth of F_2.

(1) tt and Tt

(2) Tt and tt

(3) Tt and Tt

(4) tt and tt

61) Which of the following sets of statements regarding Thalassemia and Sickle-cell anaemia is **correct**?

(1) Thalassemia is a qualitative problem of synthesising an incorrectly functioning globin, while Sickle-cell anaemia is a quantitative problem of synthesising too few globin molecules.

(2) alpha-Thalassemia is controlled by a single gene HBA on chromosome 11, while beta-Thalassemia is controlled by genes HBB1 and HBB2 on chromosome 16.

(3) Sickle-cell anaemia is caused by the substitution of Glutamic acid by Valine at the sixth position of the beta globin chain.

(4) In Thalassemia, the defect is always due to a point mutation in the gene coding for the alpha chain only.

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62) Observe the cross given here :-

♀ \ ♂	Ab	aB	AB	ab
?	AAbb	AaBb	AABb	Aabb
?	Aabb	aaBb	AaBb	aabb

What is the genotype of crossed plants ?

(1) AaBb × AAbb

(2) AAbb × aaBB

(3) AaBb × Aabb

(4) AaBb × AaBb

63) A man of A blood group marries a woman of AB blood group. Which type of progeny would indicate that man is heterozygous A:-

(1) AB

(2) A

(3) O

(4) B

64) In a plant species, flower colour yellow is dominant over white and fruit shape round is dominant over elongated. Crossing was performed between two pure lines one having yellow flower and round fruit and another with white flower and elongated fruit. About 20 plants survived in F_1 progeny. Plant of F_1 were allowed to self fertilize and about 960 plants survived in F_2. If the trait follow Mendelian inheritance the number of plants would have yellow flower and round fruit in F_1 and F_2 are respectively :-

(1) 20, 960

(2) 20, 540

(3) 10, 180

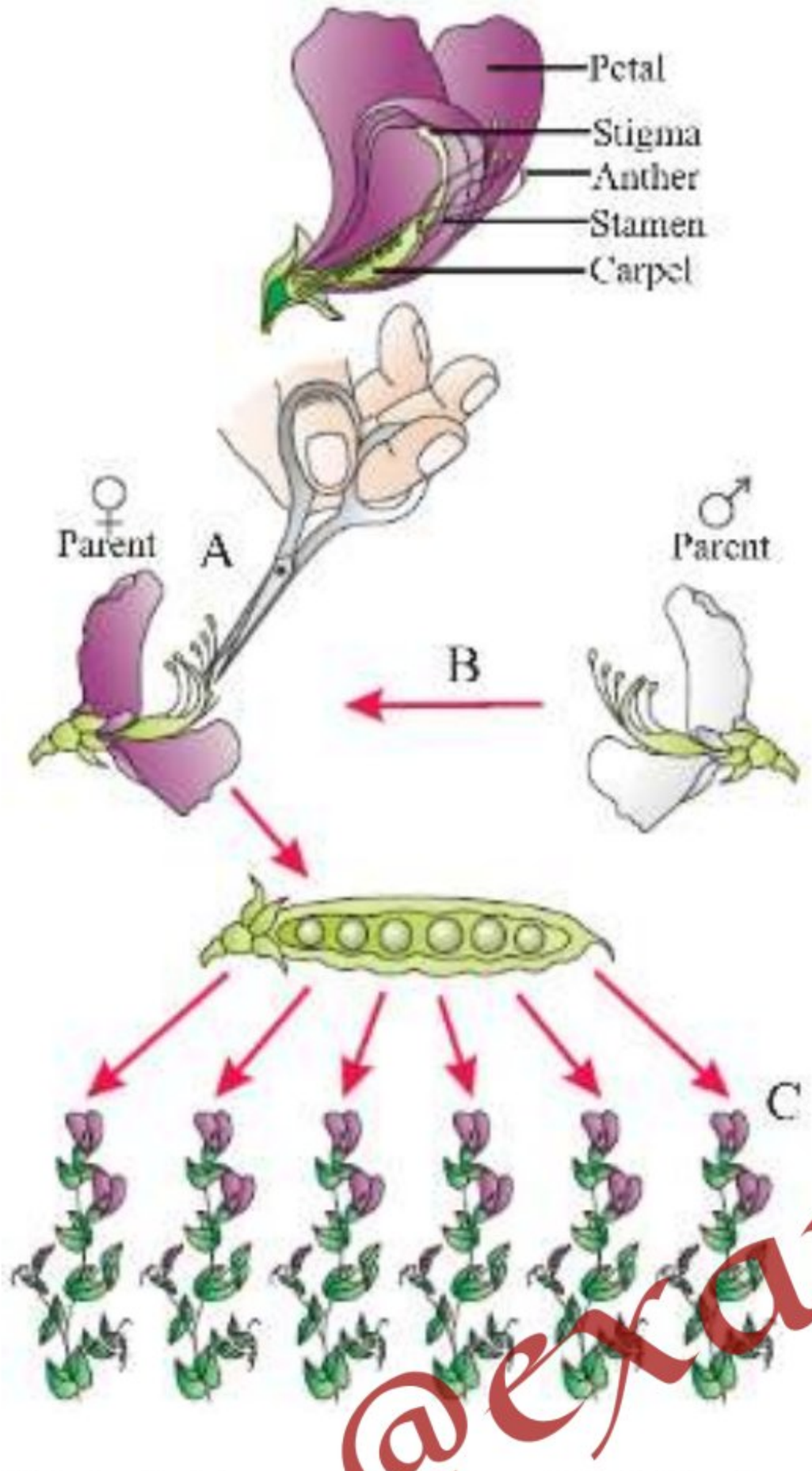
(4) 10, 60

65) The figure given below is of experimental technique of Mendel. Select the option giving **correct** identification of A, B & C :-

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(1) A-Artificial cross pollination; B-Hybrid offsprings; C-Emasculation.

(2) A-Emasculation; B-Bagging & tagging; C-Hybrid offsprings.

(3) A-Emasculation; B-Artificial cross pollination; C-Hybrid offsprings.

(4) A-Bagging & tagging; B-Emasculation; C-Hybrid offsprings.

66) F_1 wild female was crossed with recessive male and progeny was produced as follows :

++/yw = 202
yw/yw = 198
+w/yw = 798
y+/yw = 802

Now match correct combination :-

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	Type of inheritance	Strength of linkage	Distance between y and w	Type of arrangement
(1)	Maximum crossing over	80%	20 cM	Cis
(2)	Incomplete linkage	90%	10 cM	Cis
(3)	Incomplete linkage	80%	20 cM	Cis
(4)	Incomplete linkage	80%	20 cM	Trans

(1) 1

(2) 2

(3) 3

(4) 4

67) Down Syndrome is most commonly caused by trisomy 21, a condition resulting from a failure of chromosomes to segregate properly during cell division. Statistical and genetic tracking shows that the vast majority of trisomy 21 cases arise from a nondisjunction event during maternal meiosis I. What is the physical state of the extra chromosome 21 in a child with Down Syndrome resulting from a maternal meiosis I error?

(1) Two identical copies of a single maternal sister chromatid.

(2) Two different maternal homologous chromosomes plus one paternal chromosome.

(3) Two paternal sister chromatids plus one maternal copy.

(4) A single chromosome 21 that has undergone spontaneous duplication during fertilization.

68) A diploid organisms is heterozygous for 5 loci and homozygous for 2 loci. Out of 5 heterozygous loci 3 are competency linked, then how many types of gametes will be formed:

(1) 8

(2) 12

(3) 14

(4) 28

69) **Assertion(A)**: In honey bee population, sons do not have fathers but have grandfathers.
Reason(R): Haploidy in drones of honey bee is due to parthenogenesis while fertilisation results into female bees.

(1) Both (A) and (R) are True and (R) is the correct explanation of (A).

(2) Both (A) and (R) are True but (R) is not the correct explanation of (A).

(3) (A) is True but (R) is False.

(4) (A) is False but (R) is True.

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70) Mendel selected fourteen true breeding plant varieties, as pairs, which were similar except for one character with contrasting traits. How many dihybrid crosses were possible with these characters?

(1) 7

(2) 14

(3) 21

(4) 35

71) Which of the following statements are correct?
1. Pedigree analysis can predict inheritance probability
2. Mutation always decreases fitness
3. Recombination increases genetic variation
4. Genetic disorders are always dominant

(1) 1 and 3 only

(2) 1, 2 and 4

(3) 2 and 3 only

(4) 1, 2, 3 and 4

72) A geneticist observed an interesting pattern in the inheritance of a trait in a family with the following observations:
1. In the 1st generation, the mother was affected.
2. None of the sons were unaffected in the 2nd generation.
3. All 4 siblings were normal in the 3rd generation (including one son).
4. A son in the 4th generation was affected despite both parents being normal.

Identify the correct mode of inheritance for this trait:

(1) Autosome-linked Dominant

(2) Allosome-linked Dominant

(3) Autosome-linked Recessive

(4) Allosome-linked Recessive

73) Two genes, **L** and **M**, show **16% recombination**. If **1000** offspring are produced in a test cross, how many will be **recombinants**?

(1) 80

(2) 160

(3) 320

(4) 500

74) In a pea plants, green pod colour is dominant over yellow pods. 1000 seeds taken from a pea plant on germination produces 760 green pod and 240 yellow pod plants. The parental genotype and phenotype of the seed plant are

(1) Heterozygous and yellow

(2) Heterozygous and green

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(3) Homozygous and yellow

(4) Homozygous and green

75) The colour of fruits is controlled by two genes. If a fruit having 3 and 4 dominant alleles show 25x and 30x colour respectively, then what will be the minimum phenotype?

(1) x

(2) 5x

(3) 10x

(4) 15x

76) Consider the following statements regarding alleles-
i. Alleles are different forms of a gene
ii. Alleles occupy some locus on homologous chromosome.
iii. Allele is a variant of the sequence of nucleotides at a particular locus on a DNA molecules.
How many statements is or are wrong?

(1) 0

(2) 3

(3) 2

(4) 1

77) Consider the following four statements I, II, III and IV and select the correct statement :-
I. Mendelian experiments has a large sampling size, which gave greater credibility to the data that he collected.
II. Recessive allele influences the appearance of the phenotype even in the presence of a dominant allele.
III. All the traits of a character can be found only when population studies are made.
IV. In F_2-generation of a Mendelian monohybrid cross, the tall and dwarf traits were identical to their parental types and shows blending inheritance.
The correct statements are

(1) I and III

(2) III and IV

(3) II and IV

(4) II and III

78) In Mendelian dihybrid cross, the percentage of recombinants in F_2 generation is :-

(1) 37.2%

(2) 62.8%

(3) 37.5%

(4) 62.5%

79) ABO blood groups are controlled by the gene I. The gene I has three alleles I^A, I^B & i^o. Select the incorrect statement from the following:-

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(1) The alleles I^A & I^B produces a slightly different form of sugar

(2) The allele i^o does not produce any sugar

(3) I^A & I^B are incompletely dominant over i^o allele

(4) I^A & I^B are equally dominant allels

80) The height of a plant is under control of polygenic inheritance. The plant having genotype 'aabb' has 20 cm height and height of plant having genotype AaBb is 50 cm, what will be height of a plant having AABB genotype ?

(1) 45 cm

(2) 60 cm

(3) 80 cm

(4) 15 cm

81) Which of the following is not the example of pheno copies ?

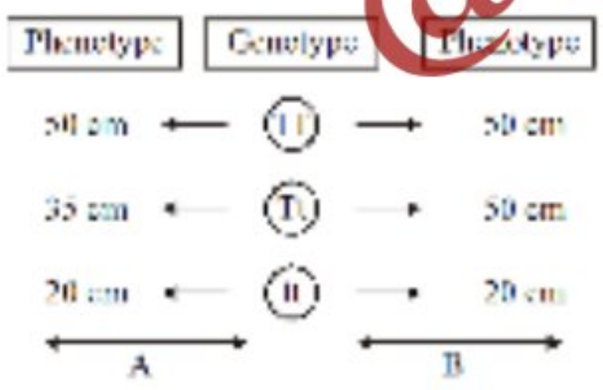
(1) TT → Tall
tt → Tall

(2) Tt → Dwarf
tt → Dwarf

(3) TT → Tall
Tt → Tall

(4) TT → Dwarf
tt → Dwarf

82)



	A	B
(1)	Polygenic qualitative	Polygenic quantitative
(2)	Monogenic quantitative	Monogenic qualitative
(3)	Monogenic quantitative	Monogenic quantitative
(4)	Monogenic qualitative	Monogenic qualitative

(1) 1

(2) 2

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(3) 3

(4) 4

83) 20 chromosome are present in diploid organism, if a chromosomal mutation occurs in this species, than choose correct option with respect to aneuploidy:

(1) 21 chromosomes with monosomy

(2) 22 chromosomes with trisomy

(3) 19 chromosomes with monosomy

(4) 21 chromosomes with tetrasomy

84) Which among the following is a true genotypic composition of a gamete:-

(1) ALLEN

(2) NEET

(3) BOTANY

(4) ZOOLOGY

85) **Statement (1):** Independent pairs segregate independently of each other.
Statement (2): One pair segregates independently of another pair.
i) Statement (1) is applicable to gene only.
ii) Statement (1) is applicable to chromosome only.
iii) Statement (1) is applicable to gene as well as chromosome.
Iv) Statement (2) is applicable to gene only.
v) Statement (2) is applicable to chromosome only.
vi) Statement (2) is applicable to gene as well as chromosome.
Select the correct option:

(1) i, ii, iii

(2) iii, iv, v

(3) iii, v

(4) iii, vi

86) **Statement I :** Dominance is not an autonomous feature of a gene in case of pleotropy.
Statement II : Most of human genetic disorders are pleotropic.

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I is incorrect but statement-II is correct.

(3) Statement-I is correct but statement-II is incorrect.

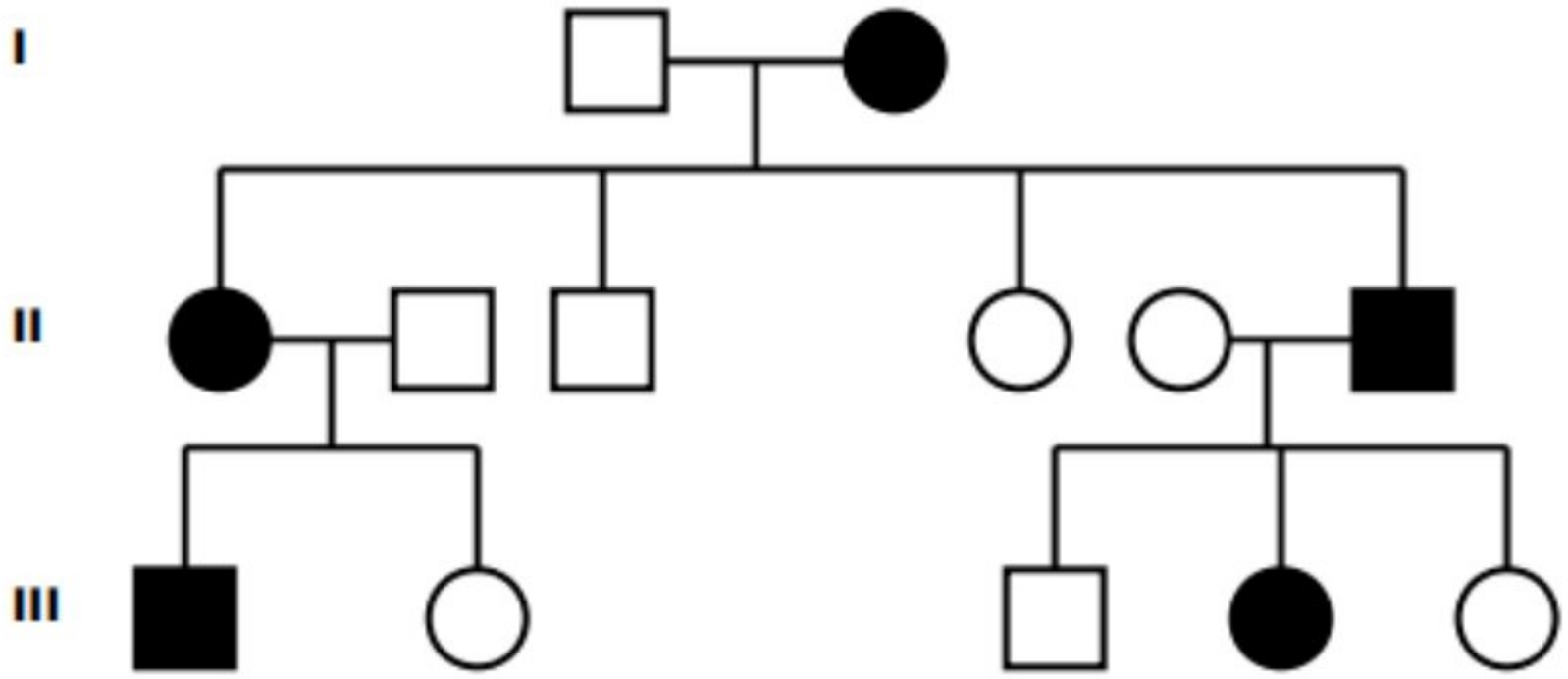
(4) Both Statement-I and statement-II are correct.

87) Pedigree analysis is best suited to study:-
(i) Mendelian disorders
(ii) Non - Mendelian disorders
(iii) All human genetic disorders
(iv) Criss-cross Inheritance
(v) Chromosomal aberrations

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(vi) Polyploidy

(1) i, ii, iii
(2) i, iv, i
(3) iv, v, vi
(4) i, iii, iv



88)
How many among the following disorders follow in the given pedigree chart-
a.) Autosomal Dominant
b.) Autosomal Recessive
c.) X-Linked Dominant
d.) X-Linked Recessive
e.) Y-Linked

(1) a, b, c
(2) a, b, d
(3) b, c, d
(4) a, b, c, d, e

89) **Read the following statements carefully:**
(I) Both β-Thalassemia and Sickle cell anaemia occur due to mutations in the same gene.
(II) Quality of Haemoglobin is compromised in Thalassemia.
(III) There is a structural abnormality in the protein formed in β-Thalassemia.
(IV) α-Thalassemia trait will best follow the independent assortment.
(V) Thalassemia is a quantitative inheritance while Sickle cell Anaemia is a qualitative inheritance.
Select the option with "Incorrect statements":

(1) I, II, III, IV
(2) II, III, IV, V
(3) I, III, IV, V
(4) I, II, IV, V

90) Which among the following statements are correct?
(i) Mutagens are mutation generators.
(ii) Most of the mutations are recessive.
(iii) Illegitimate crossing over can alter the linkage groups.
(iv) Crossing over can change the linkage groups.

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(v) Polyploidy in plants is more common than in animals.

(1) (i), (ii), (iii)

(2) (i), (ii), (iii), (iv)

(3) (i), (ii), (iii), (v)

(4) (i), (ii), (iii), (iv), (v)

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NOVA

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	3	4	1	3	3	1	2	1	3	3	3	2	3	1	3	2	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	3	1	2	4	2	2	4	2	2	4	1	4	1	3	2	3	1	4
Q.	41	42	43	44	45															
A.	1	3	2	4	1															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	2	1	2	3	4	3	4	4	4	3	3	3	4	1	4	4	4	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	1	4	1	1	3	3	2	1	2	2	2	1	4	4	4	1	4	4	4
Q.	86	87	88	89	90															
A.	3	4	4	3	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	4	4	3	4	1	2	1	1	4	1	4	4	3	2	3	3	3	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	4	4	4	4	3	3	4	1	4	1	1	2	1	4	3	2	4	4	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	1	2	2	2	1	2	4	3	2	3	4	1	2	3	3	3	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	4	2	3	4	2	1	1	3	1	4	2	2	3	1	1	3	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	2	3	3	3	4	2	1	2	3										

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NOVA

SOLUTIONS

PHYSICS

1) Conceptual

2) Conceptual

3) Conceptual

4) Conceptual

5) Conceptual

6) Conceptual

7) Conceptual

8) Conceptual

9) Conceptual

10) Conceptual

11) Conceptual

12) Conceptual

13) Conceptual

14) Conceptual

15) conceptual

16) Conceptual

17) COnceptual

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18) Conceptual

19) Conceptual

20) Cocneptual

21) Conceptual

22) Conceptual

23) Conceptual

24) Conceptual

25) Conceptual

26) Conceptual

27) Conceptual

28) Conceptual

29) Conceptual

30) Conceptual

31) Conceptual

32) Conceptual

33) Conceptual

34) Conceptual

35) Cocneptual

36) Conceptual

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37)

Conceptual

38) Conceptual

39) Conceptual

40) Conceptual

41) Conceptual

42) Conceptual

43) Conceptual

44) Conceptual

45) Cocneptual

CHEMISTRY

46)

Conceptual

47) $K_a = C\alpha^2$

$\Rightarrow 1.6 \times 10^{-5} = 0.01 \times \alpha^2 \Rightarrow \alpha = 4 \times 10^{-2}$

$$\alpha = \frac{\wedge_m}{\wedge_m^\circ} \Rightarrow 0.04 = \frac{\wedge_m}{380 \times 10^{-4}}$$

$$\Rightarrow \quad \wedge_m = 15.2 \times 10^{-4} S\,m^2/mol$$

$$\Rightarrow \quad \wedge_m = 15.2\ S\,cm^2/mol$$

$G^* = 0.01\ m^{-1} = 10^{-4}\ cm^{-1}$

As we know

$$\wedge_m = \frac{K \times 1000}{M} \Rightarrow 15.2 = \frac{K \times 1000}{0.01}$$

$K = 15.2 \times 10^{-5}\ S\ cm^{-1}$

As we know $K = G \times G^*$

$$G = \frac{15.2 \times 10^{-5}}{10^{-4}} = 1.52\,S$$

48) The limiting molar conductivity of H^+ is very high as compared to K^+ due to smaller size and

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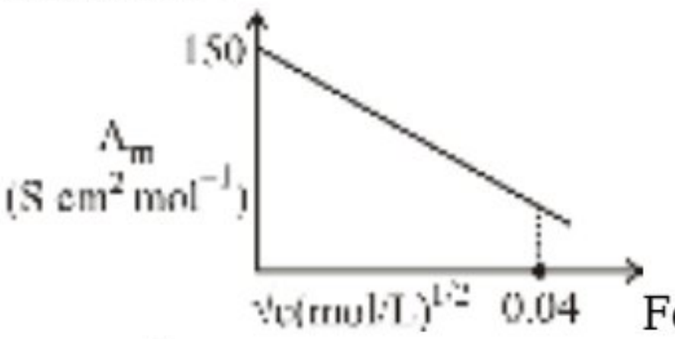
thus higher velocity.

49)

Conceptual

50)

Explain the Question : Find the limiting molar conductance $\left(\Lambda_m^0\right)$ of KCl from the graph of Λ_m versus $\sqrt{C}$ at 298 K.
Concept : This question is based on Debye-huckel-onsager equation
Solution :



For a strong electrolyte:

$\Lambda_m = \Lambda_m^0 - b\sqrt{C}$

The limiting molar conductance Λ_m^0 is obtained by extrapolating the straight line to $\sqrt{C} = 0$.
From the graph, the intercept on the Λ_m axis at $\sqrt{C} = 0$ is 150 S cm^2 mol^{-1}.
Final answer : 2

51)

Conceptual

52) **Explain the Question :** The problem involves calculating K_{sp} of $BaSO_4$ using its conductivity and equivalent
conductance.
Concept : This question is based on $K_{sp} = S^2$, where S = solubility (mol/L).
Equivalent conductance (Λ_{eq}) and conductivity (K) are related as:
$\Lambda_{eq} = (K \times 1000)/N$, where N = normality = solubility × valency factor.
Solution :
Given : $K = 3.06 \times 10^{-6}\ \Omega^{-1}cm^{-1}$, $\Lambda_{eq} = 1.53\ \Omega^{-1}cm^2\ eq^{-1}$
$\Lambda_{eq} = (K \times 1000)/N$
$1.53 = (3.06 \times 10^{-6} \times 1000)/N$
$N = 2 \times 10^{-3}$ N
Molarity = N/2 = 1×10^{-3} mol/L
$K_{sp} = S^2 = (10^{-3})^2 = 10^{-6}$
Final answer : 4

53)

Conceptual

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54) **Question Explanation:**
Find the correct statements for electrolysis of aqueous NaCl solution.

Concept:
This question is based on electrolysis of brine (chlor-alkali process).

Solution:
Cathode:
$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$
Anode: $2Cl^- \rightarrow Cl_2 + 2e^-$
OH^- ions remain in solution forming NaOH, so the solution becomes basic:
• pH > 7
• Turns red litmus blue
• Products are H_2 and Cl_2

Hence, all statements are correct.

Final Answer: Option (4)

55)

Conceptual

56)

Question Explanation: Factors affecting the life of a Daniell cell.

Concept: This question is based on the life of a galvanic cell depends on how long the electrodes can participate in redox reactions.

Solution:

A larger electrode offers a greater surface area and more material for reaction, thus increasing cell life.
In a Daniell cell (Zn | Zn^{2+} || Cu^{2+} | Cu), Zn is the anode and Cu is the cathode. Increasing the size of **zinc electrode and Cu^{2+} ions** extends the duration before it is consumed or fully deposited.
Final Answer: Option 4

57)

Question Explanation: Identify correct statements for Daniel cell (Zn–Cu cell).
Concept: This question is based on

A. In Daniel cell: Zn (anode) → $Zn^{2+} + 2e^-$, $Cu^{2+} + 2e^- \rightarrow$ Cu (cathode).
B. Electrons flow from anode to cathode, while current flows from cathode to anode externally.
C. Anode is negative, cathode is positive.
D. Salt bridge maintains charge balance; ions have nearly equal mobility.
E. Cation concentration decreases in anodic compartment and increases in cathodic

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compartment.

Solution:

(i) Current flows from cathode to anode → **True**
(ii) Anode is negatively polarized → **True**
(iii) Mobility of cation and anion in salt bridge ≈ same → **True**
(iv) Cation concentration in anodic compartment decreases → **False** (it increases)
(v) Cation concentration in cathodic compartment decreases → **True**
4 statements are correct
Final Answer: Option 3

58)

Conceptual

59)

$$E_{cell} = \frac{0-0.059}{h}\log\frac{[P]}{[R]}$$

Anode $H_{2(g)} \rightarrow 2H^+ + 2e^-$

$$\frac{2H^+ + 2e \rightarrow H_2}{H_2(P_1) + 2H^+(C_2) \rightarrow 2H^+(C_1) + H_2(P_2)}$$

$$E_{cell} = \frac{-0.059}{2}\log\frac{C_1^{\,2} \times P_2}{C_2^{\,2} \times P_1}$$

$\because P_1 = P_2 = 1$ atm

$$E_{cell} = \frac{0.059}{2}\log\frac{C_2^{\,2}}{C_1^{\,2}}$$

$C_2 > C_1$ Ecell = +ve .

60)

Conceptual

61) **Question Explanation:** Identify which redox reaction listed is non-spontaneous under standard conditions.
Concept: A reaction is non-spontaneous if its standard cell potential ($E°_{cell}$) is negative.
Solution: To overall reaction potential is
$E°_{cell} = E°_{reduction} - E°_{oxidation}$
A reaction is non-spontaneous if $E°_{cell} < 0$
(1) $Ni^{2+} + Cu \rightarrow Ni + Cu^{2+}$

A. Reduction : $Ni^{2+} \rightarrow Ni$ ($E° = -0.25$ V)
B. Oxidation : $Cu \rightarrow Cu^{2+}$ ($E° = +0.34$ V)
C. $E°_{cell} = E°_{Ni^{2+}/Ni} - E°_{Cu^{2+}/Cu}$

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(2) $2Ag+ + Cu \rightarrow 2Ag + Cu^{2+}$

$E°_{cell} = E°_{Ag+Ag} - E°_{Cu^{2+}/Cu} = 0.8V - 0.34V = +0.46\ V$ (spontaneous)

(3) $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

$E°_{cell} = E°_{Cu^{2+}/Cu} - E°_{Zn^{2+}Zn} = 0.34V - (-0.76V) = +1.10\ V$ (spontaneous)

(4) $2H^{+} + Zn \rightarrow H_2 + Zn^{2+}$

$E°_{cell} = E°_{H^{+}/H2} - E°_{Zn2+Zn} = 0.00V - (-0.76V) = +0.76\ V$ (spontaneous)

Final answer: option (1)

62) **Question Explanation:** Find the X and Y pair that makes the given redox reaction spontaneous.

Concept: Reaction spontaneity requires positive cell potential $E°_{cell} > 0$

Solution: The reaction $X + Y^{+2} \rightarrow X^{+2} + Y$ has X oxidized and Y^{2+} reduced.

$E° = E°_{Y^{2+}/Y} - E°_{X^{2+}/X}$

For spontaneity ($E°cell > 0$), the SRP of Y must be greater than the SRP of X.

checking options:

SRP $\rightarrow Zn^{+2} < Fe^{+2} < Ni^{+2}$

Final answer: option (4)

63) Efficiency $= \frac{\Delta G}{\Delta H} \times 100$

$\Delta G = -nFE$

64)

Question Explanation: We must find log K_{sp} of AgI using given electrode potentials.

Concept: $\Delta G° = -nFE°_{cell}$, $\Delta G° = -2.303\ RT \log K_{sp}$

Solution:

Step 1: Write half-reactions

Given:

$Ag + I^{-} \rightarrow AgI + e^{-}$; $E° = +0.152\ V$

Reverse it:

$AgI + e^{-} \rightarrow Ag + I^{-}$; $E° = -0.152\ V$

Also given:

$Ag \rightarrow Ag^{+} + e^{-}$; $E° = -0.800\ V$

Step 2: Add to obtain dissolution reaction

$AgI(s) \rightarrow Ag^{+} + I^{-}$

$E°_{cell} = -0.152 - 0.800$

$E°_{cell} = -0.952\ V$

Step 3: Apply equation

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n = 1

$\log K_{sp} = (nFE°_{cell}) / (2.303\ RT)$
$= (1 \times 96485 \times -0.952) / (2.303 \times 8.314 \times 298)$
$= -16.13$

Final Answer: Option 4

65)

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75)

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76)

Formation of 0.2 mol of AgCl from 0.1 mol of the complex shows that there are two ionizable Cl-atoms. Formula of the complex is $[Co(NH_3)_5Cl]Cl_2$, i.e., 1 : 2 type electrolyte.

$[Co(NH_3)_5Cl]Cl_2 \rightleftharpoons [Co(NH_3)_5Cl]^{2+} + 2Cl^-$

77) Co^{3+} $3d^6$ in SFL $t2g^6eg°$ - hence diamagnetic

78)

Conceptual

79) **Explain Question:** Limitation of VBT for colour.

Concept: VBT Limitation.

Solution: VBT cannot explain colour; it is explained by CFT. So A false, R true.

Final Answer: Option (4)

80)

Conceptual

81)

Question Explanation: The question is testing the definition of organometallic compounds, which require a direct metal-to-carbon (M-C) bond.

Concept Used: Organometallic Compounds

Answer Explanation:

A. **Ferrocene:** $Fe(\eta^5\text{-}C_5H_5)_2$ contains direct Fe-C bonds. (Organometallic)

B. **Zeise's salt:** $K[PtCl_3(\eta^2\text{-}C_2H_4)]$ contains a direct Pt-C bond. (Organometallic)

C. **Grignard reagent:** RMgX contains a direct Mg-C sigma bond. (Organometallic)

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D. **Cis-platin:** cis-$[PtCl_2(NH_3)_2]$ contains Pt-Cl and Pt-N bonds, but no metal-carbon bond. (Not organometallic)

Final Answer: Option (4)

82)

Conceptual

83)

Question explanation: The question is asking which statement correctly identifies the metal present in an important biological coordination compound.

Concept: Many biomolecules contain metal–ligand complexes: chlorophyll (Mg), hemoglobin (Fe), vitamin B_{12} (Co), enzymes like carboxypeptidase (Zn).

Solution: Check each statement:

A. Chlorophyll contains Mg, not Fe → incorrect.

B. Vitamin B_{12} contains Co, not Mg → incorrect.

C. Hemoglobin contains Fe, not Co → incorrect.

D. Carboxypeptidase contains Zn as the metal → correct.

Final answer: (Option 4).

84)

Explanation – The question asks which of the given options results in a blue coloration.

Concept – (Various reaction & the colour product formed)

$CuSO_4(aq) + 2NH_4OH(aq) \rightarrow Cu(OH)_2(s) + (NH4)_2SO_4(aq)$

pale blue

$4FeCl_3 + 3K_4[Fe(CN)_6] \rightarrow Fe_4[Fe(CN)_6] + 12KCl$

deep blue pigment

$CuSO_4(s) + 6H2O(l) \rightarrow [Cu(H_2O)_6]SO_4(aq)$

light blue color

Final Answer – Option (4)

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85)

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86)

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87)

Question Explanation: The question asks why AgBr dissolves in hypo solution (sodium thiosulphate, $Na_2S_2O_3$), i.e., which complex ion is formed that increases solubility.

Concept:

A. Hypo ($S_2O_3^{2-}$) forms stable complex ions with Ag^+.
B. Formation of a stable complex reduces free Ag^+ concentration, shifting equilibrium toward dissolution of AgBr.

Solution:
AgBr is sparingly soluble in water:
$AgBr(s) \rightleftharpoons Ag^+ + Br^-$
In presence of excess thiosulphate ions:
$Ag^+ + 2S_2O_3^{2-} \rightarrow [Ag(S_2O_3)_2]^{3-}$
Because $[Ag(S_2O_3)_2]^{3-}$ is highly stable, Ag^+ ions are removed from solution, so more AgBr dissolves.

Final Answer: Option (4)

88)

Conceptual

89)

Question Explain:
The incorrect statement regarding the properties of Prussian blue.

Concept:
Coordination chemistry, magnetic moments, homoleptic complexes, Prussian blue structure.
Formula:
•Magnetic moment (μ) = $\sqrt{[n(n+2)]}$ BM, where n is the number of unpaired electrons.

Solution:
In Prussian blue complex $Fe_4[Fe(CN)_6]_3$ outer ion is Fe^{+3} ($3Cl^5$), so it have 5 unpaired electron, so 3rd option is wrong.

Final Answer: Option (3)

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90) NCERT XII, Unit 9 (2017), Pg # 248, Part-I
Weak ligand (F^-) ⇒ No pairing

BIOLOGY

91)

Explain Question : Identify the four statements (a–d) and select the option which includes all the incorrect ones only.

Concept : Male reproductive system

Solution : The correct option is (2).

(a) Incorrect — the number "750 tubules per lobule" is wrong (each lobule has only 1–3 seminiferous tubules).

(b) Incorrect — Sertoli cells are supportive cells and do not undergo meiosis.

(c) Correct — interstitial spaces do contain Leydig cells and blood vessels.

(d) Correct — scrotum helps maintain lower temperature for spermatogenesis.

Final answer : (2) Statements (a) and (b).

92) NCERT(XII) Pg# 50 Para:3.4

93) NCERT XII Pg.#50

94)

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96) NCERT XII Pg. 48

97)

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98)

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99)

Conceptual

100) **Explain Question** : identify correctness of each statement and count incorrect ones.
Concept : Human Reproduction
Solution : The correct answer is option 4. All four statements are incorrect.

A. (A) Incorrect → Hymen is not a reliable indicator of virginity.The hymen is not a reliable indicator of virginity as it can be broken or stretched due to various non-sexual activities.

(B) Incorrect → Acrosome contains enzymes, not mitochondria (mitochondria are in middle piece).(C) Incorrect → After fertilization, corpus luteum persists, it does not transform into Graafian follicle.(D) Incorrect → Delivery of foetus is called parturition, not gestation (gestation = pregnancy period).
Final Answer : 4

101)

Conceptual

102) Correct Answer: (4)

103) NCERT Pg # 30

104)

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105)

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106)

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107)

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108)

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110)

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111)

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112) Module 1, Pg.# 40

113)

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114)

Option 4 is the correct answer.

A. **(A) Fusion of male & female gamete** - This process is known as **fertilization**, and it typically occurs in the **ampulla** of the fallopian tube. So, A matches with (iii).

B. **(B) Attachment of blastocyst** - The blastocyst, a hollow ball of cells, attaches to the lining of the uterus, specifically the **fundus of the uterus**. So, B matches with (i).

C. **(C) Reactivation of sperms** - Sperms are reactivated in the **vagina**. So, C matches with (iv).

D. **(D) Development of Graafian follicle** - Graafian follicles develop in the **ovary**. So, D matches with (ii).

115) NCERT-XII Pg#27,35

116) NCERT Pg. # 61

117)

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118)

NCERT Pg. 47

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119)

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120)

Conceptual

121) NCERT (XII) ; Page No. # 33, 34, 35

122)

Question explanation: Identify the functions of extra-embryonic membranes in humans and compare with other vertebrates.

Concept: Extra-embryonic membranes

Solution: The correct answer is option 1:

A. A – Allantois: Correct – in humans, it does not store excretory wastes.

B. B – Reptiles and birds: Correct – in these animals, allantois stores excretory wastes.

C. C – Amnion: Correct – it keeps the foetus suspended in amniotic fluid.

D. D – Yolk sac: Correct – it helps in early blood cell formation (haematopoiesis).

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So, all blanks are correctly filled as Allantois, Reptiles and birds, Amnion, and Yolk sac.

Final answer: option 1

123)

Question explanation: Identify the correct extra-embryonic membranes involved in implantation and development.

Concept: Extra-embryonic membranes

Solution: The correct answer is option 2:

A. A – Chorion: Correct – it gives rise to chorionic villi after implantation.

B. B – Uterine endometrium: Correct – chorionic villi embed into the uterine lining for exchange.

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C. C – Allantois: Correct – it forms blood vessels connecting embryo with placenta.

D. D – Yolk sac: Correct – it reduces after 8 weeks of development.

So, all blanks are correctly filled as Chorion, Uterine endometrium, Allantois, and Yolk sac.

Final answer: option 2

124)

Question explanation: Identify the correct functions of extra-embryonic membranes to fill the blanks.

Concept: Extra-embryonic membranes

Solution: The correct answer is option 1:

A. A – Amnion: Correct – it secretes amniotic fluid.

B. B – Desiccation: Correct – amniotic fluid prevents drying (desiccation) of the foetus.

C. C – Chorion: Correct – it forms chorionic villi for implantation and exchange.

D. D – Yolk sac: Correct – it is the first haematopoietic organ.

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So, all blanks are correctly filled as Amnion, Desiccation, Chorion, and Yolk sac.

Final answer: option 1

125)

Conceptual

126) NCERT-XII, Pg. # 51

127)

Conceptual

128) NCERT-XII, Pg. # 38, Para-2.7

129) NCERT XII Pg # 27,29,32

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130)

Question Explanation: Identifying events that do not occur during pregnancy.

Concept : Pregnancy

Solution : The correct answer is option 3.

(A) Decrease in progesterone → Incorrect, because progesterone remains high to maintain endometrium.

(B) Degeneration of corpus luteum → Incorrect, corpus luteum is maintained (by hCG) in early pregnancy.

(C) Stoppage of menstrual cycle → Correct, this occurs during pregnancy.

(D) Endometrial hyperplasia and myometrium atrophy → Incorrect, because myometrium does not atrophy, it enlarges(Hypertrophy)

(E) Increase in maternal thyroxine → Correct, mild rise occurs to support higher metabolic demand.

A. Events that don't occur during pregnancy: A, B, and D

Final answer : 3

131)

Conceptual

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132)

Question Explanation: The question evaluates the correctness of two statements related to human reproduction.

Concept: Human Reproduction

Solution:

A. Statement A is correct - Humans reproduce sexually and give birth to young ones, hence are viviparous.

B. Statement B is correct - Cyclic menstruation indicates a normal reproductive phase in females and occurs from menarche to menopause.

Final answer: option - 1

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133)

Question Explanation: Identify the reason why testes are located outside the abdominal cavity in the scrotum.
Concept: Male reproductive system - scrotum
Solution: The correct answer is **Option 1**.

A. **Assertion (A):** In humans, testes are situated outside the abdominal cavity within the scrotum.
    A. This is **true**. The testes are located in the scrotum, a pouch of skin that hangs outside the body.
B. **Reason (R):** The scrotum helps in maintaining the low temperature of the testes necessary for spermatogenesis.
    A. This is also **true**. Sperm production (spermatogenesis) requires a temperature slightly lower than the normal body temperature. The scrotum helps to maintain this lower temperature by being located outside the body.

    C. Both the assertion and the reason are **true**, and the reason correctly explains the assertion.

Final Answer: Option 1

134) NCERT Page no. 47 - 48

135) **Concept :** Location and physiological significance of the scrotum.
Solution:

A. Statement A is correct: "The testes are situated outside the abdominal cavity within a pouch called scrotum."
B. Statement B is incorrect: states that the low temperature (2–2.5°C lower than the normal internal body temperature) is "necessary for spermatogenesis" (the process of sperm formation). It does not attribute this temperature requirement specifically to *spermiation* (the release of sperms from seminiferous tubules) or the *secretion of testosterone* (which is hormonal and occurs within the interstitial cells).
C. Option 1 is incorrect because Statement B is false.
D. Option 3 is incorrect because Statement A is true.
E. Option 4 is incorrect because Statement A is true.

Final Answer: Option(2)

136) Ans. (2)
2 is the correct answer

137)

Conceptual

138)

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Explain Question : The question asks to determine the genotypes and phenotypes of the F_1 (first filial) generation and the F_2 (second filial) generation.

Concept : This question is based on History of Mendel

Solution :
His work was not supported with required data because Mendel's work was meticulously conducted and supported by extensive experimental data gathered from his pea plant experiments. The issue was more about the lack of understanding and acceptance of his quantitative approach by the scientific community at the time.

Final Answer : option (2)

139)

Conceptual

140) **The correct answer is:**
3. Law of Segregation

A. When this individual undergoes sexual reproduction, it will produce two types of gametes: one with allele T and one with allele t. Therefore, the chances of a gamete being Tare 50%, and the chances of it being tare also 50%.

Now, the question asks about the chances that the T allele from the pollen grain will pollinate an egg of genotype T. Since the pollen grain has a 50% chance of carrying the T allele, the answer is 50%.

141)

Explain Question: What is the term for graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross,

Concept: Punnet square (Monohybrid Cross)

Solution:

Graphical representation to calculate the probability of all possible genotypes of offspring in a genetic cross, is known as Punnet Square.

So, the correct answer is: 2.

Final answer: Option 2

142)

Conceptual

143)

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144) Correct Answer: (1)

145)

Conceptual

146)

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147)

Conceptual

148)

Solution:
Gene A follows complete dominance:
• AA Aa (dominant phenotype) → 3 parts
• aa (recessive phenotype) → 3 parts
• Ratio = 3 : 1
Since the two genes segregate independently we multiply the two ratios : (3:1) × (1: 2: 1) = 3 : 6: 3 :1 : 2 : 1
Correct Option: (3)

149) **Solution:** It is sex-linked recessive. Carrier mother (XX^c) passes X^c to 50% of sons. 8% males, 0.4% females.
Final Answer : (3)

150) Genotype of P_1 and one fourth of F_2 will be tt and tt.

151) **Solution:** Sickle-cell anaemia is caused by the substitution of Glutamic acid by Valine at the sixth position of the beta globin chain. Option A reverses the definitions (Thalassemia is quantitative, Sickle-cell is qualitative). Option B cites incorrect chromosomes/genes.
Final Answer : (3)

152)

Conceptual

153) B

154)

Conceptual

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155) NCERT XII Pg.# 71

156)

Conceptual

157) During Meiosis I, homologous chromosome pairs (one maternal, one paternal) must separate. If nondisjunction happens to chromosome 21 during maternal Meiosis I, both homologous chromosomes travel together into the egg cell instead of separating. When fertilized by a normal sperm containing one paternal chromosome 21, the zygote ends up with two different maternal homologous chromosomes and one paternal chromosome.

158)

Conceptual

159) Conceptual.

160) Correct Answer: Option 3 (21)
Explanation: Total plant varieties = 14
Total characters (pairs) = 14 / 2 = 7

A dihybrid cross involves selecting 2 characters at a time out of 7.
Number of possible dihybrid crosses = 7C2 = (7 × 6) / (2 × 1) = 21

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161) **Question Explanation:** The question asks about concepts of pedigree analysis, mutation, recombination, and genetic disorders.

Concept: Variation & inheritance

Solution:
Pedigree analysis helps predict inheritance probability (correct).
Mutation does not always decrease fitness (incorrect).
Recombination increases variation (correct).
Genetic disorders are not always dominant (incorrect).

Final Answer: Option 1. 1 and 3 only

162) **Correct Answer: (D) Allosome-linked Recessive**

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Generation	Observed Phenotype	Explanation
Generation I	Mother affected	Mother is homozygous recessive (X^aX^a).
Generation II	None of the sons were unaffected	Sons inherit X^a from the affected mother, making all sons affected (X^aY).
Generation III	All 4 siblings normal	The affected father (X^aY) passes X^a to his daughters (making them carriers, X^AX^a) and Y to his sons (making them normal, X^AY).
Generation IV	Son affected from normal parents	The carrier mother from Gen-III (X^AX^a) passes X^a to her son, making him affected (X^aY).

163) **Solution:** Recombinants = 16% of 1000 = **160 individuals**.
Final Answer : (2)

164)

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165)

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168)

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169)

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170)

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Conceptual

171) **Explain Question –** Which of the following is not an example of phenocopy.

Concept – This question based on phenocopy.

Solution –
Phenocopy occurs when an environmental condition causes an individual to show a trait typically determined by genes, but without the Corresponding genotype.
different genotype show the same phenotype → phenocopy

Final Answer – Option (3)

172)

Conceptual

173)

Ncert Pg No: 91 2021-22

174) Conceptual

175)

NCERT-XII, Pg. # 74

176) Both Statement-I and statement-II are correct.

177) Conceptual

178) **Explanation: Autosomal Dominant (a):** Direct NCERT example of Myotonic Dystrophy. The trait does not skip generations.
Autosomal Recessive (b): Genotypically possible if unaffected parents are heterozygous carriers (Aa).
X-Linked Dominant (c): Possible because an affected heterozygous mother (X^DX^d) can pass the trait to both sons and daughters.
X-Linked Recessive (d) & Y-Linked (e): Incorrect because unaffected sons are born to an affected mother (rules out X-recessive), and females cannot show Y-linked traits.

179) Conceptual

180) Conceptual

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